2. Bill of installation tools

NO.	NAME	SPECIFICATION	PATTERN
1	Electric wrench	M5/M8/M10/M12/M14/M16/M18	
2	Screwdriver	samll size	
3	Leveling instrument	1m	
4	Allen wrench	6mm (for M8)	
5	Open-end wrench	8/13/16/18/21/24/27mm	
6	Rubber hammer	medium size	
7	Tape measure	10m	
8	Construction line	100m	
9	Angle gauge	±90°	
10	Torque wrench	Torque Range: 0–450 N·m	
11	Long nose pliers	4in	

info@esetsolar.com +86 516-83152579 www.esetsolar.com

3. Schematic diagram of the finished installation product

The installation effect diagram is shown in Figure 3.1:

1. *The following model diagram is for reference only.

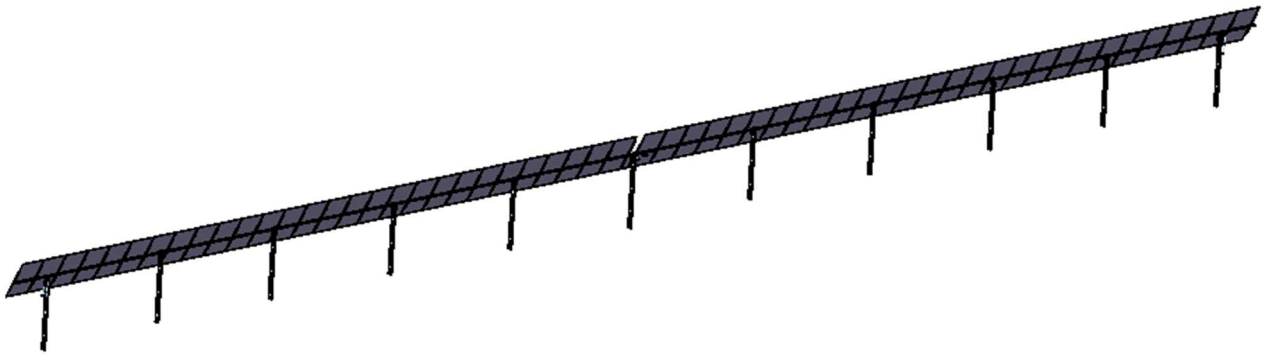


Figure 3-1 Overall effect diagram of the solar tracker for reference.

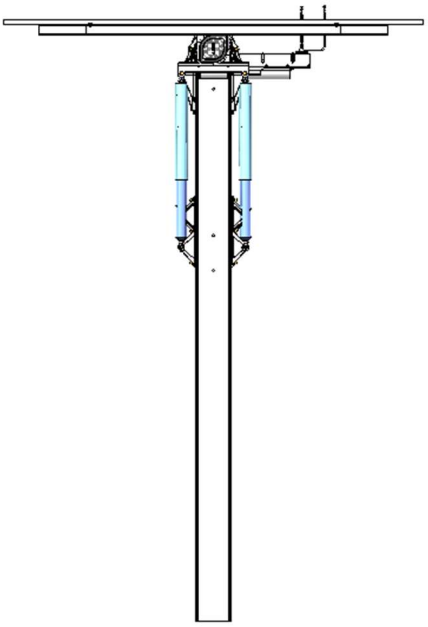
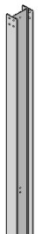
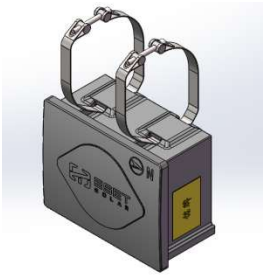
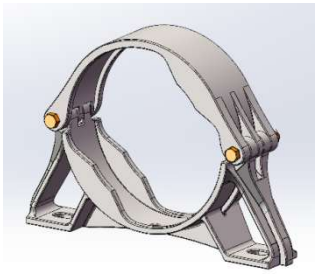
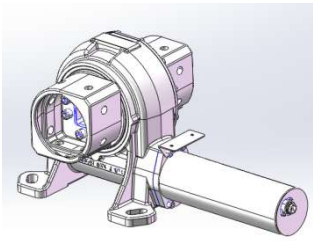
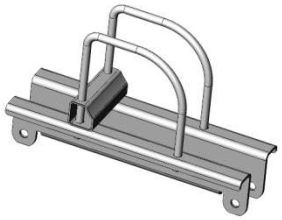
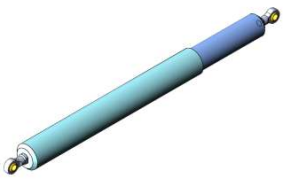
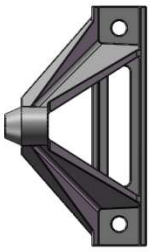


Figure 3-2 Side view effect diagram of the solar tracker for reference.

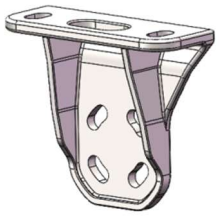
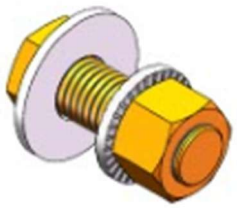
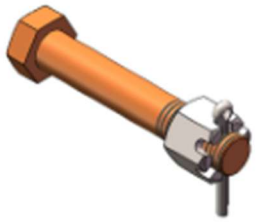
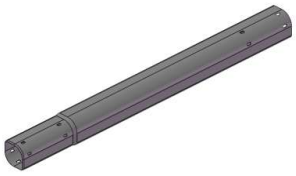
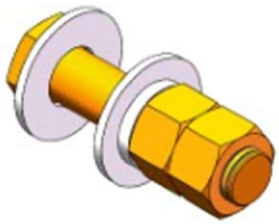
info@esetsolar.com +86 516-83152579 www.esetsolar.com

4. List of major components

Table 4-1 List of major components for reference.

NO	Description	Schematic	NO	Description	Schematic
1	Post		10	TCU BOX	
2	Bearing housing		11	Slewing reducer	
3	Bearing base		12	Polymer bearing wear parts	
4	Damper hoop		13	Damper	
5	Damper lower fixture		14	wireless receiver	

info@esetsolar.com +86 516-83152579 www.esetsolar.com

6	Reducer base		15	M12/M14 Bolts with hole	
7	Torque tube connecting the reducer		16	M14 Assembly for Bearing housing	
8	Connecting torque tube		17	M16 Assembly for Reducer base & Bearing base	
9	Purlin		18	M18 Assembly for Reducer	

For detail tightening torque of the bolts, please check Appendix 1.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

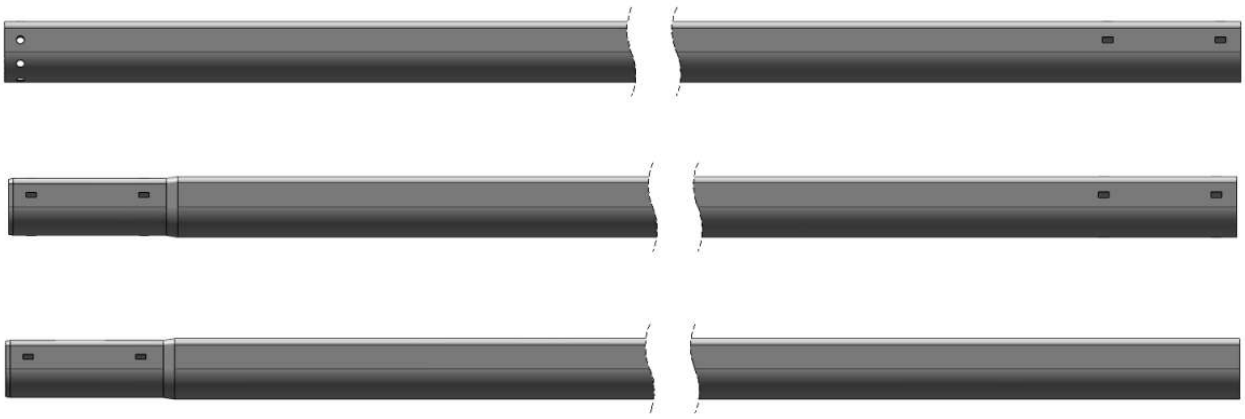
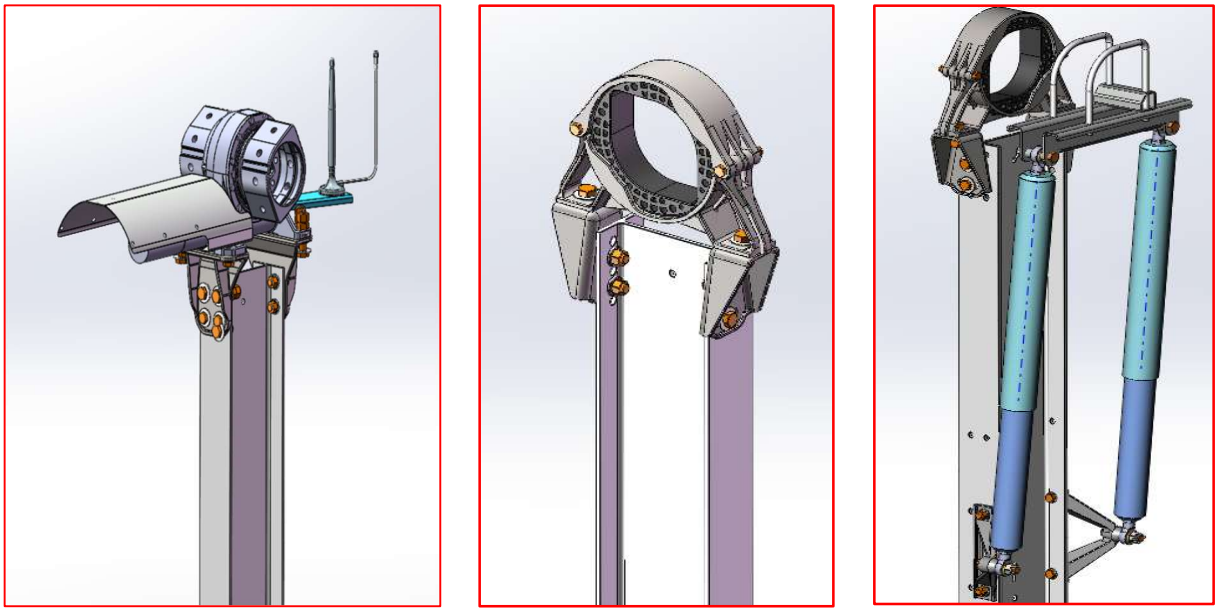


Figure 4-2 Drawing of 3 types of torque tube



A. Driving post B. Non-driving post C. Damper post

Figure 4-3 Schematic effect drawing of the three types of post

info@esetsolar.com +86 516-83152579 www.esetsolar.com



Figure 5.1-5: Foundation post central reference line

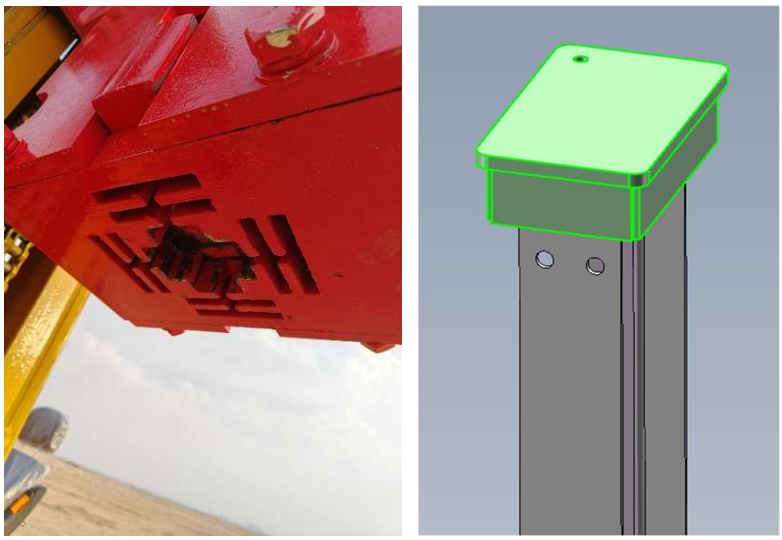


Figure 5.1-6: Foundation pile cap image

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.1.2 Foundation layout

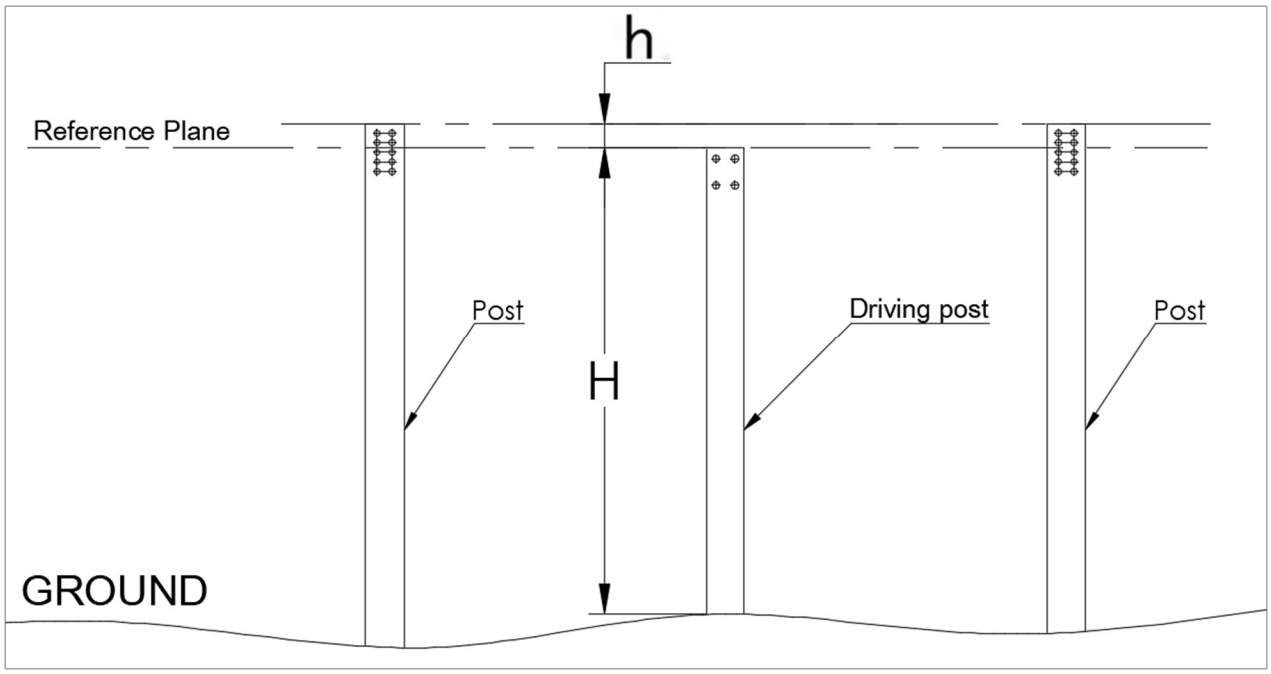


Figure 5.1.2-1: Schematic diagram of foundation depth and height from the ground

Remark:

① All above dimensions identified as “H”, “h”,and numbers of general posts are project specific.Please refer to **Appendix 2.1:Installation of posts**.

② For details about the post distribution refer to the foundation drawings to obtain specific foundation depths and distances to ram.

③ This installation manual takes the top of the posts as reference. The top of each Non-Driving Post must be h refer to **Appendix 2.1:Installation of posts**. higher than that of the Driving Posts. The post is always vertically installed no matter if there is slope or not.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.1.3 Terrain requirement

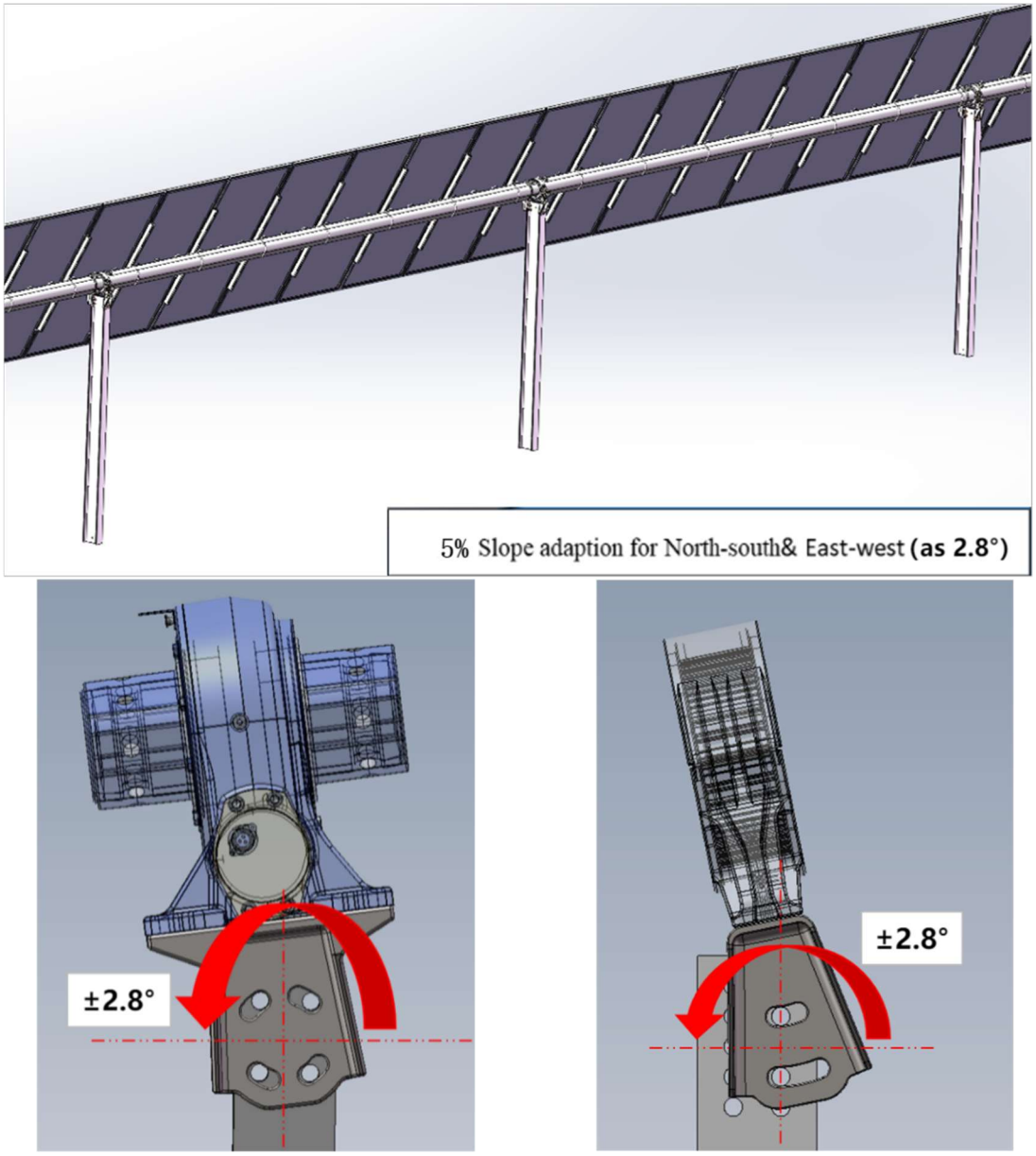


Figure 5.1.3-1: Schematic diagram of pile foundation slope adaption

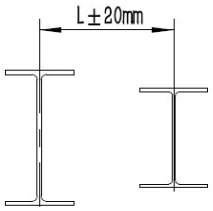
(≤5% for North-south & East-west slope)

info@esetsolar.com +86 516-83152579 www.esetsolar.com

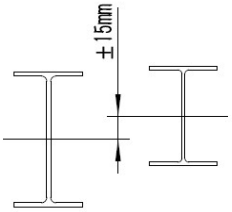
5.1.4 Allowable tolerance of foundation



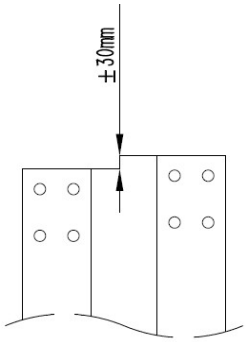
Torsional angle



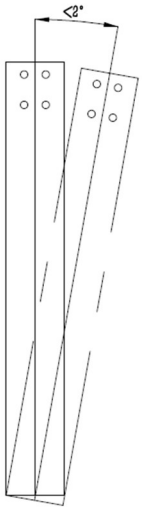
North-south direction span



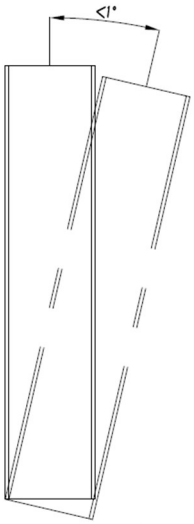
East-west direction span



Height difference



N-S Verticality



W-E Verticality

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.2 Installation of posts (Step 2)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: leveling instrument, and related equipment
3. Embed the posts into the ground according to the designed embedment depth, and measure the levelness and verticality with a leveling instrument;

Remarks:

① The installation direction of posts is the north and south;
② Refer to **Appendix 2.1:Installation of posts** for details on foundation spacing and height requirements.;
③ Refer to **Appendix 2.1:Installation of posts** for post installation location and spacing.
④ *Take the following model as an example for reference.

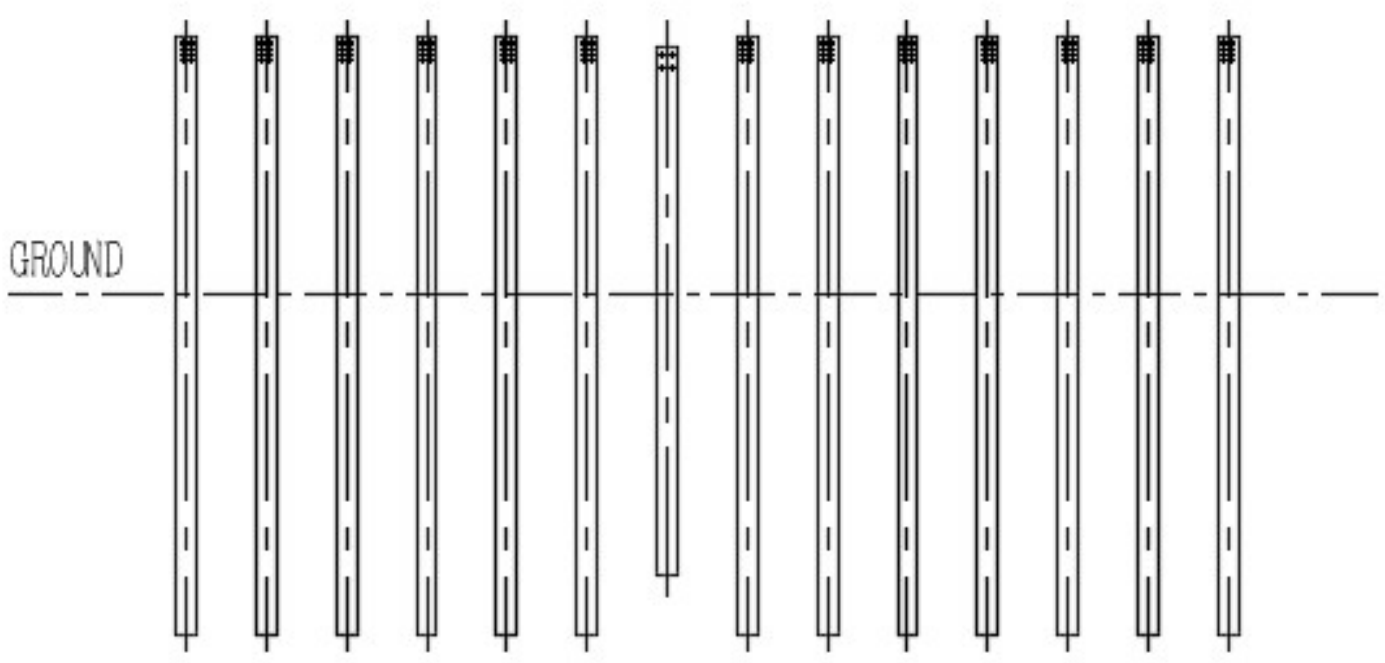


Figure 5.2-1: Schematic diagram of pile foundation spacing.

For detail information, please check Pile foundation construction guidance.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.3 Installation of reducer base and bearing base (Step 3)

1. Prepare the reducer base mounting parts and bearing base mounting parts, and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 21mm open wrench * 2, 24mm open wrench * 2, leveling instrument, and related equipment.

Remarks:

① The plane formed by the upper surface of the reducer base shall be parallel to the elevation plane of the non-drive post.

② ☆ After installation, measure the level of the both ends of the reducer driving shaft, the tolerance of torsion angle is ≤ 0.5°.If the angle cannot be adjusted within the tolerance range, report promptly.

③ The 5 holes near the reducer side is preferred when installing the bearing base, if there is interference, can be replaced to the other side of the 5 holes, the bearing base is preferred to be installed in the second and fourth holes, if there is a height difference during installation, the adjustable range is ± 25mm, as shown in Figure 5.3-6.

④ ☆ All double-layer self-locking washers shall be installed on the nut side.

Table 5.3-1: Table of materials required for the installation of the reducer base

NO.	Description	Specification	Remarks
1	Slewing reducer base	BP064	
2	Hex bolt	M16 * 50 * 30 (1 flat washer, 1 double stack self-lock gasket, 1 nut)	Torque 145~193 N.m

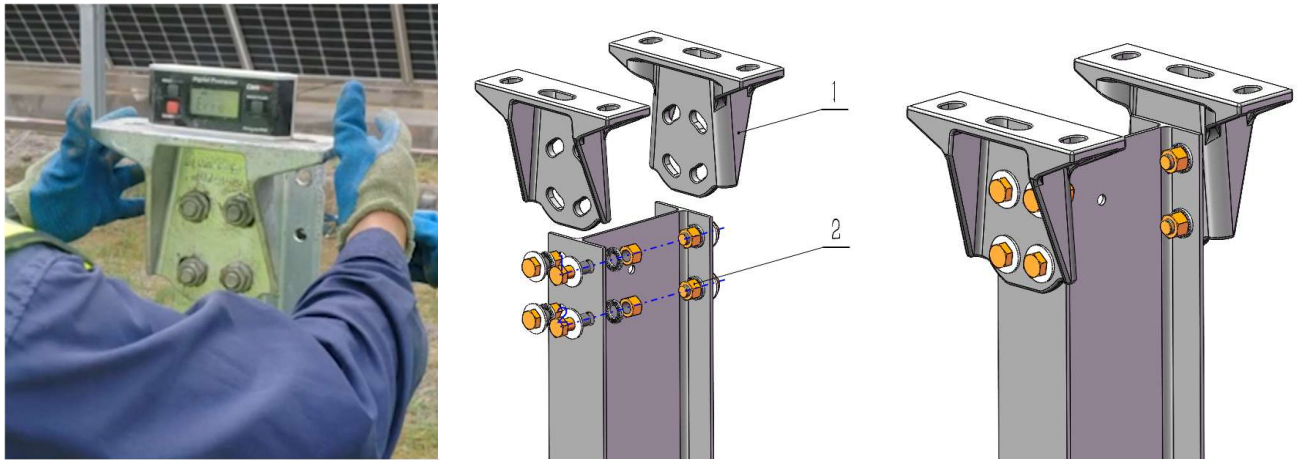


Figure 5.3-2: Installation diagram of reducer base

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Table 5.3-3: Table of materials required for the installation of the reducer base

NO.	Description	Specification	Remarks
1	Slewing reducer base	BP088	
2	Hex bolt	M14 * 40 * 30 (1 flat washer, 1 double stack self-lock gasket, 1 nut)	Torque 93~124 N.m

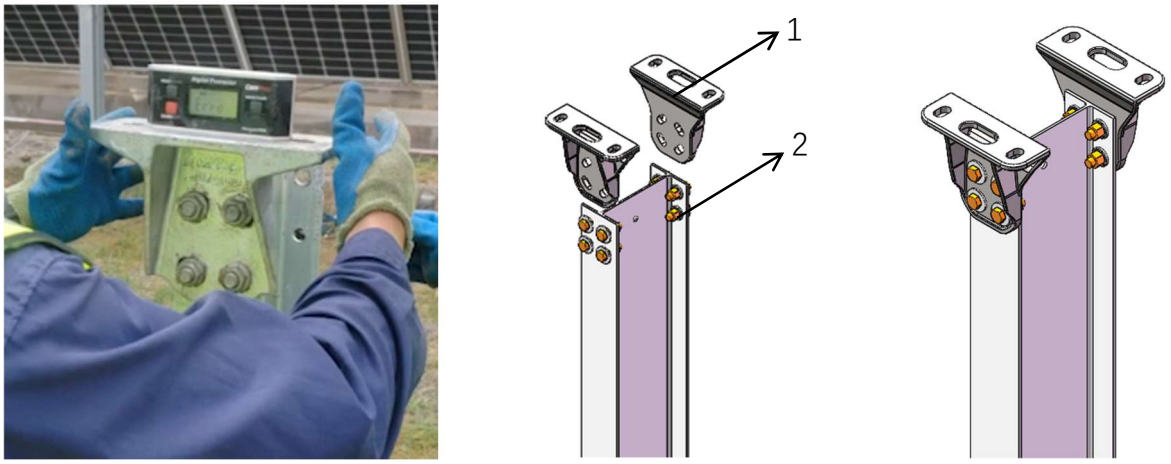
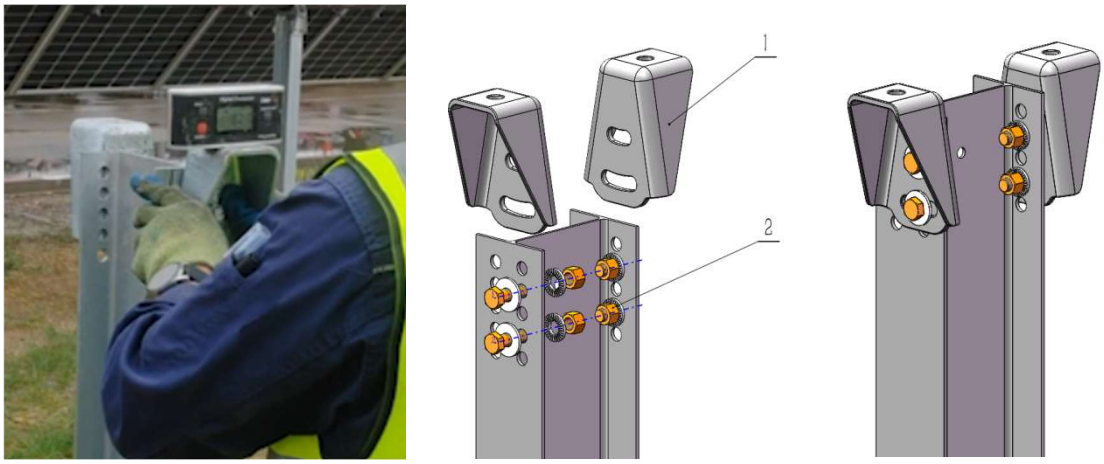


Figure 5.3-4: Installation diagram of reducer base

Table 5.3-5 Materials required for bearing base installation

NO.	Description	Specifications	Remarks
1	Bearing base	BP036H	
2	Hex bolt	M14 * 40 * 30 (1 flat washer, 1 double stack self-lock gasket, 1 nut)	Torque 93~124 N.m



info@esetsolar.com +86 516-83152579 www.esetsolar.com



Figure 5.3-6: Schematic diagram of the double stack self-lock gasket installation

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.4 Installation of slewing reducer (Step 4)

1. Prepare reducer installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 27mm open wrench * 2, leveling instrument, and related equipment
3. Place the reducer on the reducer base (☆ motor facing east), install bolts, tighten by hand; measure and adjust the level in both directions with a leveling instrument, and tighten all bolts to the specified torque.

Table 5.4-1: Materials required for reducer installation

NO.	Description	Specifications	Remarks
1	Driving post	Refer to **Appendix 2.1: Installation of posts**	
2	Slewing reducer base	BP064 / BP088	
3	Hex bolt	M18 * 80 * 50 / M16 * 75 * 50 (2 big flat washer, 1 spring washer 2 nut)	
4	Slewing reducer		

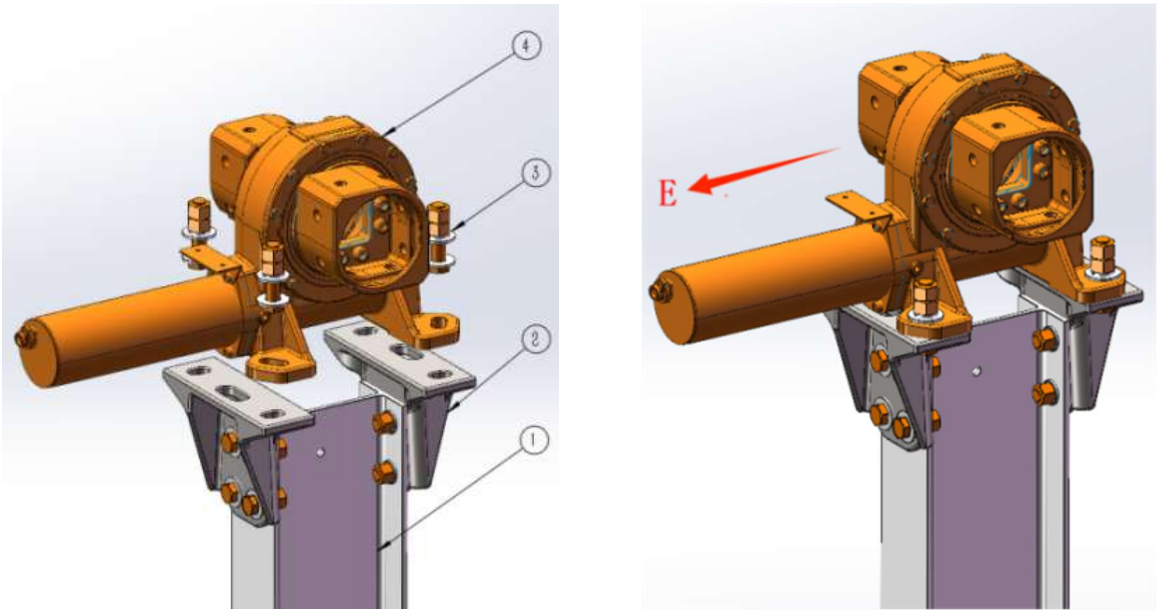


Figure 5.4-2: Explosion diagram of reducer installation

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.5 Installation of bearing housing (Step 5)

1. Prepare bearing housing parts and check the dimensions and quantity according to the product BOM and the engineering drawings;
2. Prepare installation tools: 16mm /24mm open end wrenches, and related supporting equipment.
3. Install bearing housing with bearing base and open the upper cover first;
4. After the torque tube is installed, slide the bearing onto it, push it to the position of the bearing housing until it seats into the housing, and finally secure the upper cover of the bearing housing.
5. ☆ The installation direction of the bearing housing refers to Figure 5.5-4.
6. ☆Two types of bearing housings of different sizes are compatible with different models of posts respectively.，refers to Figure 5.5-2.

Table 5.5-1: Materials required for bearing housing installation

NO.	Description	Specifications	Remarks
1	Bearing housing	BP094-200	
2	Bearing housing cover	BP090	
3	Hex bolt with pin	M10 * 90 * 26 (1 nut, 1 cotter pin 2.5 * 18)	
4	Hex bolt	M16 * 45 * 30 (1 big flat washer 1 double self-lock gasket 1 nut)	Torque 145~193 N.m
5	Bearing base	BP036H	
6	Polymer bearing	BP092	



Figure 5.5-2 bearing housing

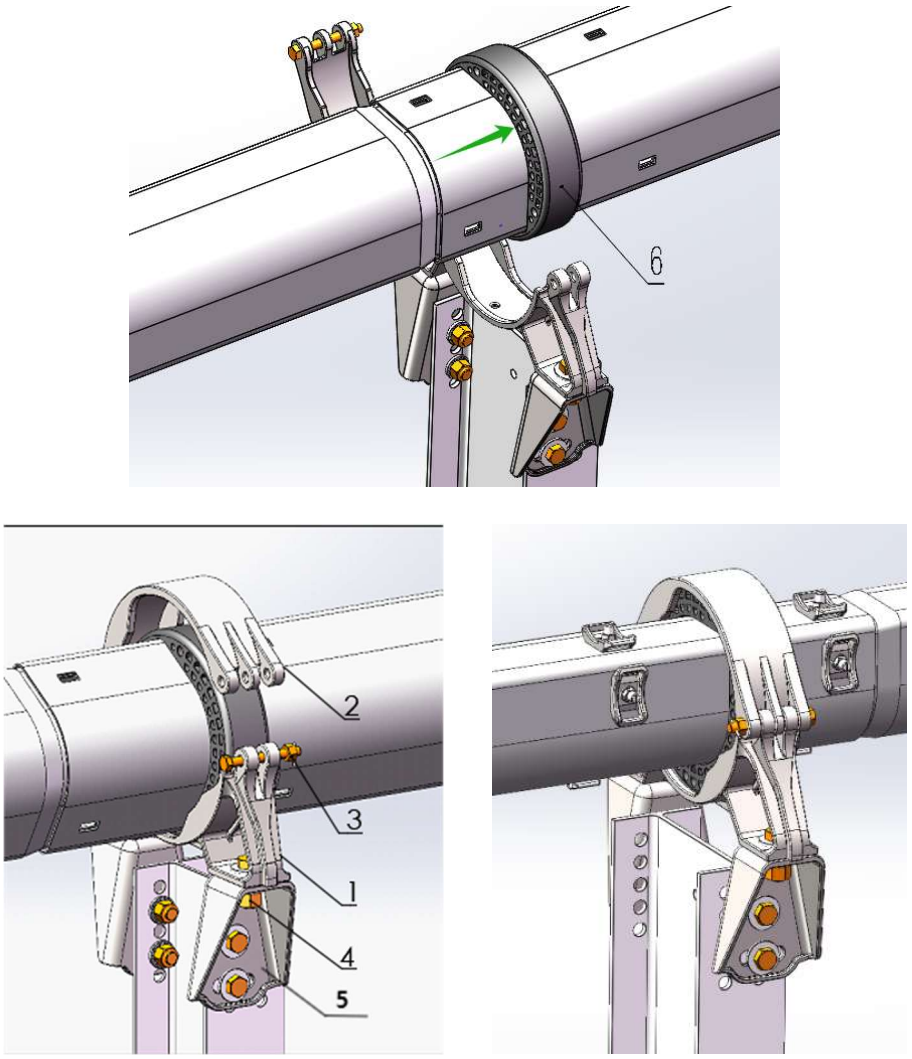


Figure 5.5-3 Installation explosion of the bearing housing

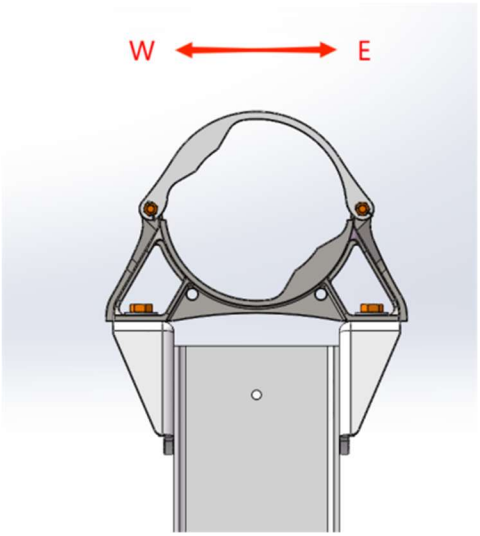


Figure 5.5-4 Mounting direction of the bearing housing

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.6 Installation of the torque tubes (Step 6)

5.6.1 Installation of the torque tubes on both sides of the reducer

1. Prepare the torque tube installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 24mm open end wrench, leveling instrument, rubber hammer, and related supporting equipment;
3. Insert the torque tube connecting the reducer from one side, install it to the reducer and tighten all bolts to the corresponding torque.

Remarks:

① When connecting the torque tube to the reducer, pay attention to the installation direction, the end with 8 φ 18 round holes will be connected to the reducer;

② The installation sequence of M16 bolts is shown in Figure 5.6.1-3;

③ ☆ The torque tube weld is installed on the lower side, a slot on the reducer shaft is left for the installation of the torque tube weld, as shown in Figure 5.6.1-4;

④ ☆ The torque tubes on both sides of reducer must be in same plane. Check the parallelism of the torque tubes on both sides with an angle instrument. After installation, the parallelism of the torque tubes on both sides of the reducer should not be greater than 0.5°, as shown in Figure 5.6.1-5;

⑤ Refer to the **Appendix 2.2 Torque tube installation process** for details on the installation locations of other torque tubes with different part numbers.;

⑥ After the installation of the solar tracker and the normal operation for three months, all the bolts connecting the torque tube should be inspected, and the bolts loosed must be tightened again.

⑦ Take the following model as an example.

Table 5.6.1-1: Materials required for torque tube installation

NO.	Description	Specifications	Remarks
1	Slewing reducer	Refer to **Appendix 5: BOM List**	
2	Torque tube reinforcing pads	BP099	
3	Torque tube	Refer to **Appendix 5: BOM List**	
4	Hex bolt	M16 * 30 * 30 (1 double stack self-lock gasket)	Torque 145~193 N.m

info@esetsolar.com +86 516-83152579 www.esetsolar.com

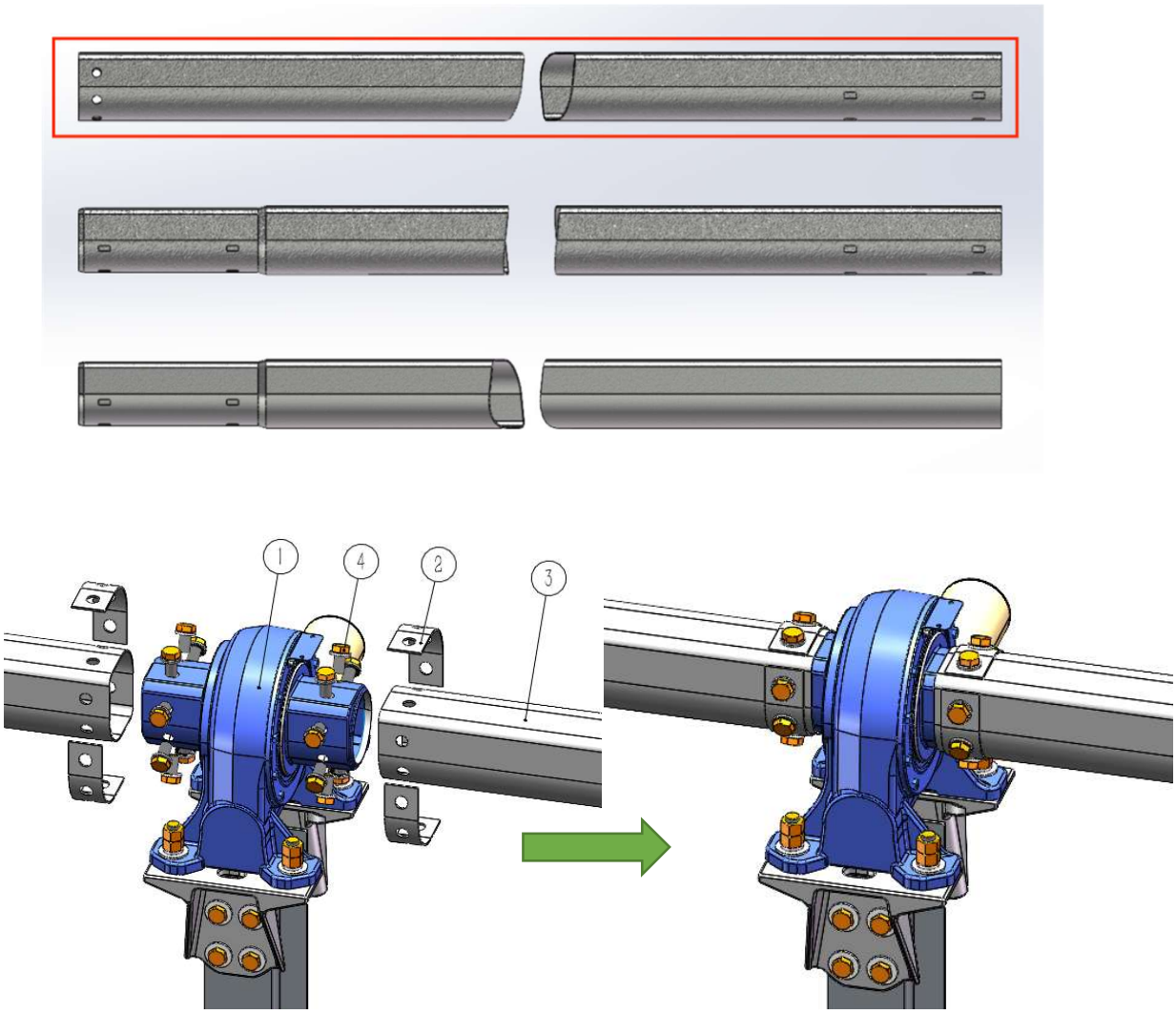


Figure 5.6.1-2: Explosion diagram and schematic diagram of the torque tube install to the reducer

info@esetsolar.com +86 516-83152579 www.esetsolar.com

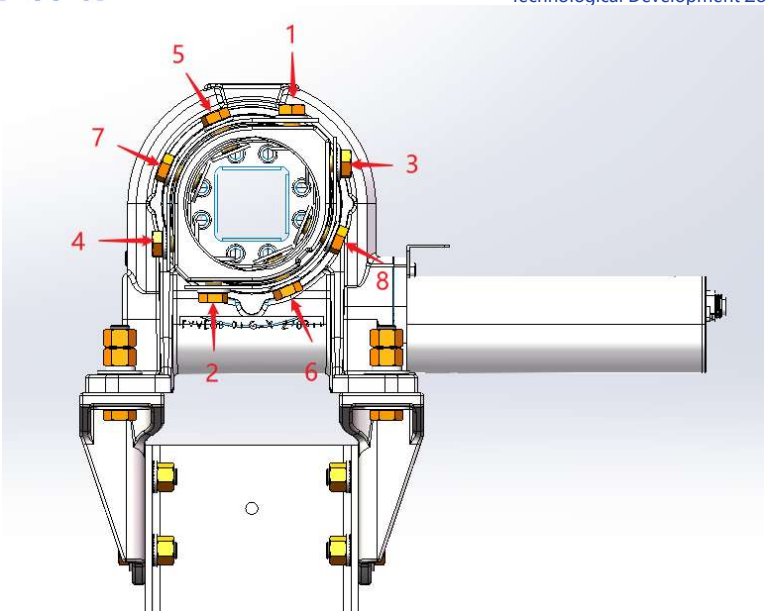
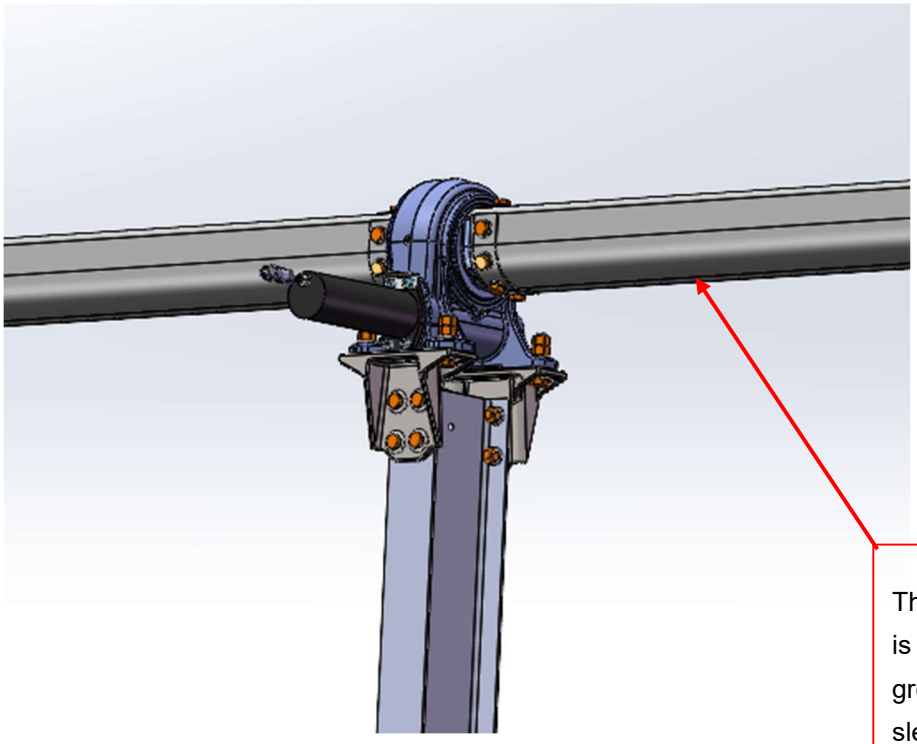


Figure 5.6.1-3: Schematic diagram of the installation sequence of the reducer bolts



The torque tube weld is installed at the groove below the slewing reducer shaft

Figure 5.6.1-4: Installation diagram of torque tube weld

info@esetsolar.com +86 516-83152579 www.esetsolar.com

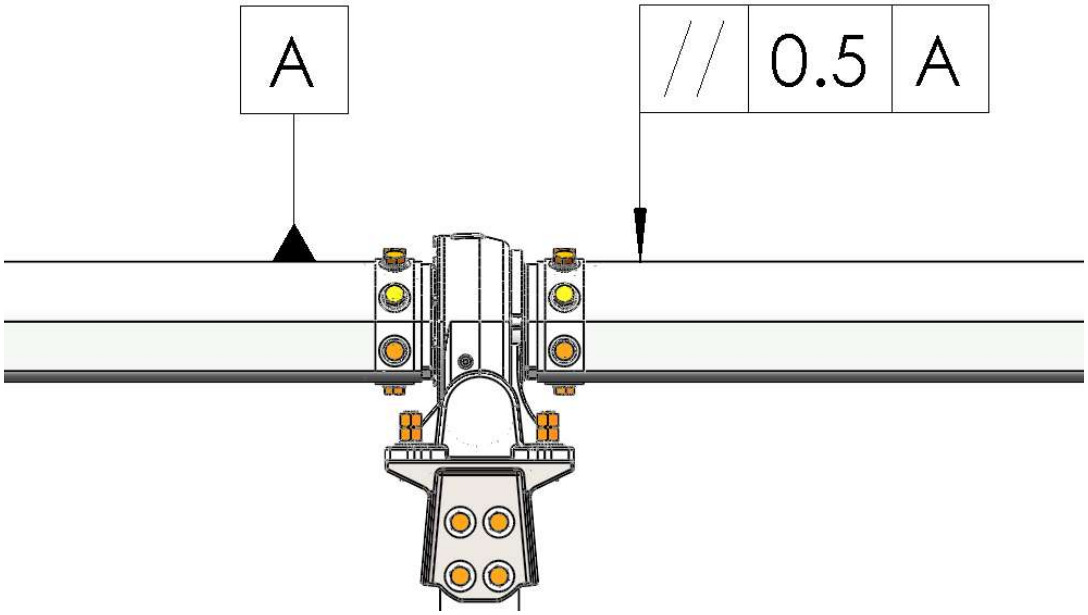


Figure 5.6.1-5: Schematic diagram of the parallelism of the torque tubes

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.6.2 Installation of diameter reduced torque tubes

1. Prepare the installation parts such as the torque tubes and carriage bolt group, and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: 16mm socket wrench, leveling instrument, rubber hammer, tape, and related equipment
3. The two torque tubes are connected by neck reduced method. Firstly, the carriage bolts are pre-installed on the shear guide gasket, then the carriage bolt group is mounted into the square holes formed by the two torque tubes, and finally tighten the carriage bolts.

Remarks:

① The installation sequence and corresponding part numbers of the torque tube refer to **Appendix 2.2:Torque tube Installation Process**.

② When installing the M10 carriage bolt, the double stacked self-locking washer and nut should be pre-installed on the bolt, then the carriage bolt group is pre-installed on the shear guide gasket, then put the T-type head bolt into the rectangular hole of the torque tube connection, tighten it clockwise, so that the T-type head bolt is stuck in the square hole of the shear guide gasket, and finally fastened with electric wrench, as shown in Figure 5.6.2-4.

③ Before installation, the '—' slot of the bolt is parallel to the front and rear block of the gasket, and be perpendicular to the front and rear block after fastening, as shown in Figure 5.6.2-5.

④ When installing the carriage bolts on the torque tube connection, tighten it twice. The installation sequence is shown in Figure 5.6.2-6.

⑤ ☆ All bolts must be fastened to the corresponding torque value.

⑥ ☆ The torque tubes on both sides of reduced torque tube must be in the same plane. Check the parallelism & torsion angle of the torque tubes on both sides with an angle instrument. After installation,the parallelism between adjacent torque tubes shall not exceed 0.5°. The cumulative torsion angle error of the entire row shall be within ±2°,as shown in Figure 5.6.2-7.

⑦ ☆ Due to the cumulative tolerance of post spacing, when interference occurs between the Bearing Housing and Carriage Bolt, the positions of the mounting holes of the Bearing Base shall be adjusted, as shown in Figure 5.6.2-8.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Table 5.6.2-1: Materials required for torque tube connection

NO.	Description	Specifications	Remarks
1	Torque tube	Refer to **Appendix 2.2:Torque tube Installation Process**	
2	Carriage bolt	MC10 * 31 * 29 (1 double-stack self-lock gasket 1 nut)	Torque 33~45 N.m

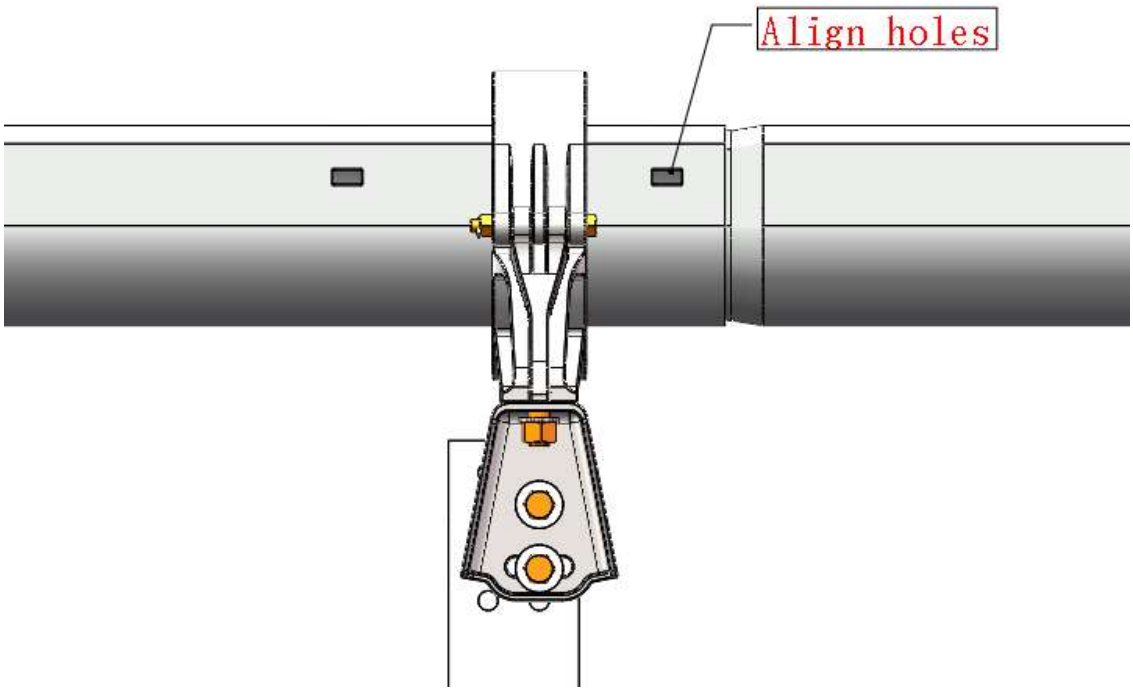


Figure 5.6.2-2: Holes position diagram of diameter reduced torque tubes

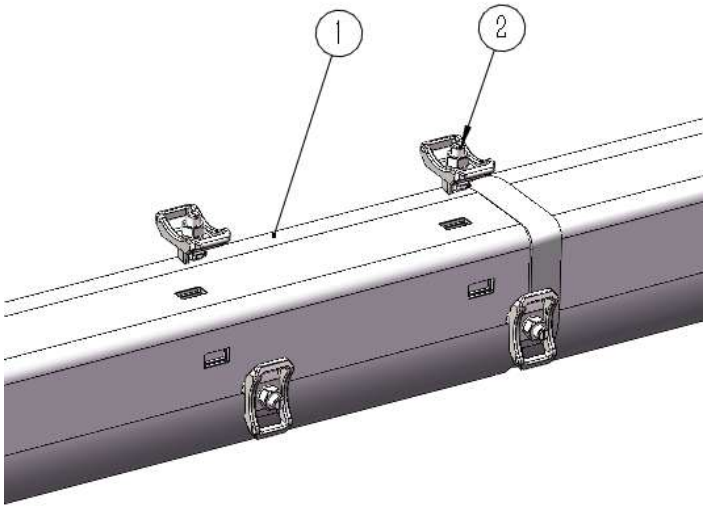


Figure 5.6.2-3: Explosion diagram of diameter reduced torque tubes installation

info@esetsolar.com +86 516-83152579 www.esetsolar.com

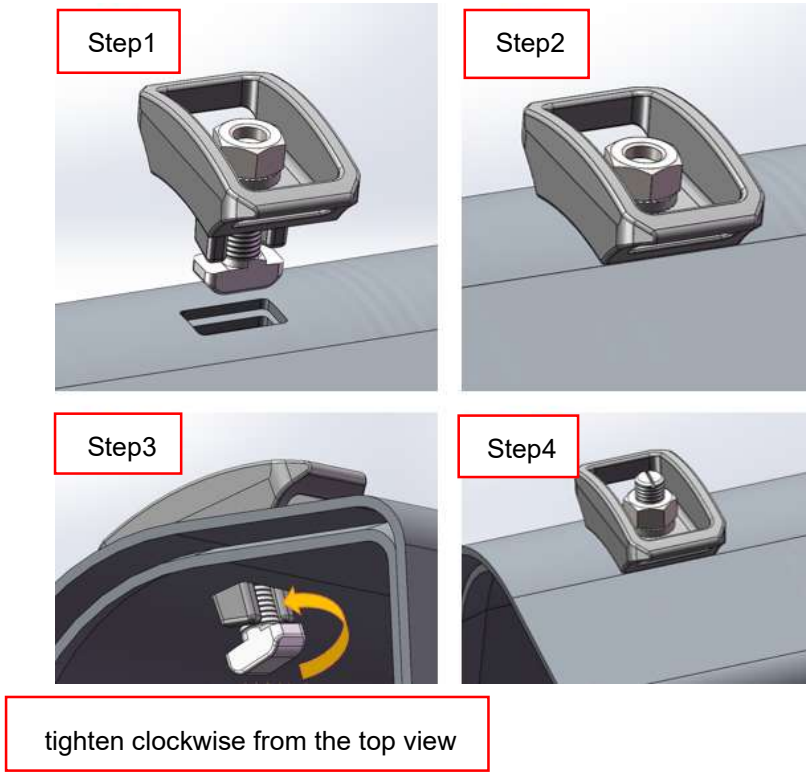


Figure 5.6.2-4: Installation diagram of carriage bolts

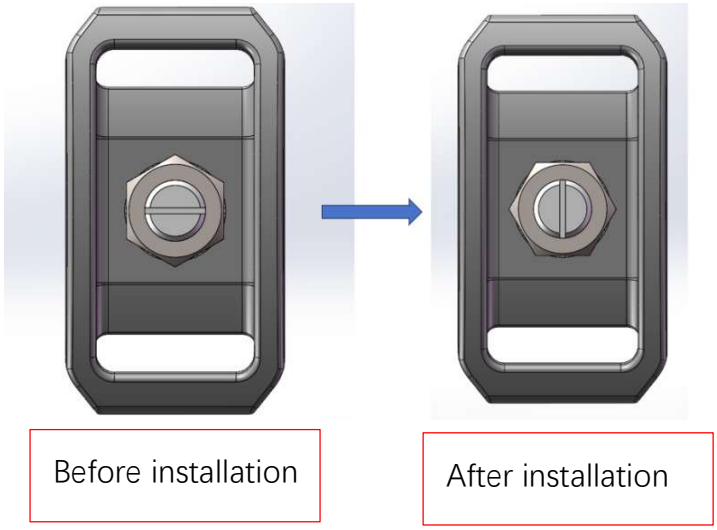


Figure 5.6.2-5: Installation diagram of carriage bolts

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Table 5.6.2-6: Materials required for torque tube connection for 1Px70

NO.	Description	Specifications	Remarks
1	Torque tube	S156	
2	Torque tube hoop	-4.5xS156	
3	Hoop spacers	-3x37	
4	Hex bolt	M12 * 50 * 40 (1 flat washer 1 double self-lock gasket 1 nut)	
5	Carriage bolt	MC12 * 35 * 30 (1 big flat washer 1 spring washer 1 nut) Big flat washer: -3x ϕ 30x ϕ 13.5	

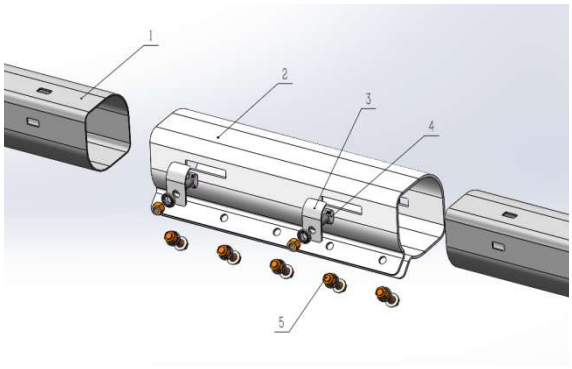


Figure 5.6.2-7.: Explosion diagram of torque tubes installation with hoop

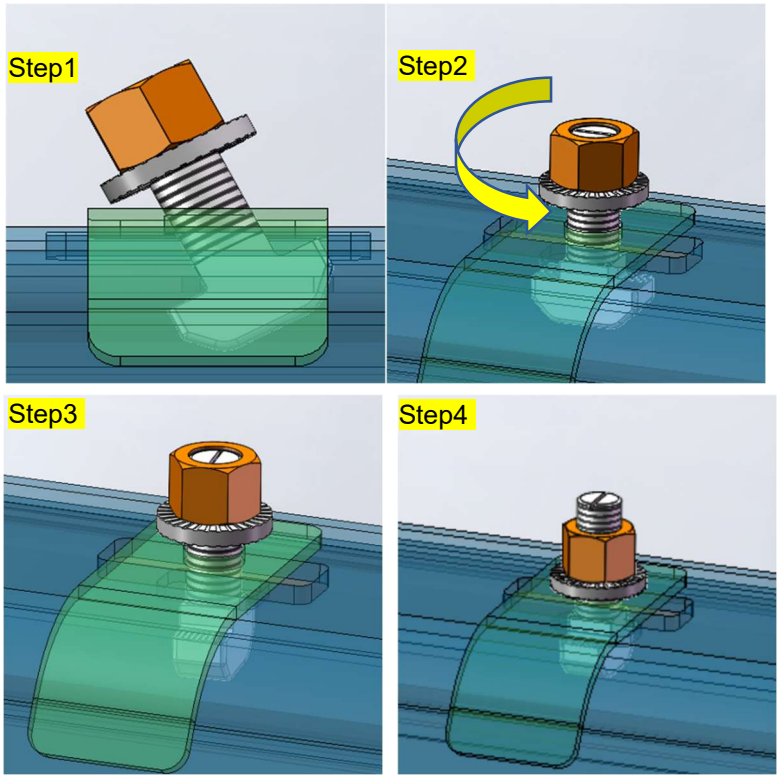


Figure 5.6.2-8.: Installation diagram of carriage bolts

info@esetsolar.com +86 516-83152579 www.esetsolar.com

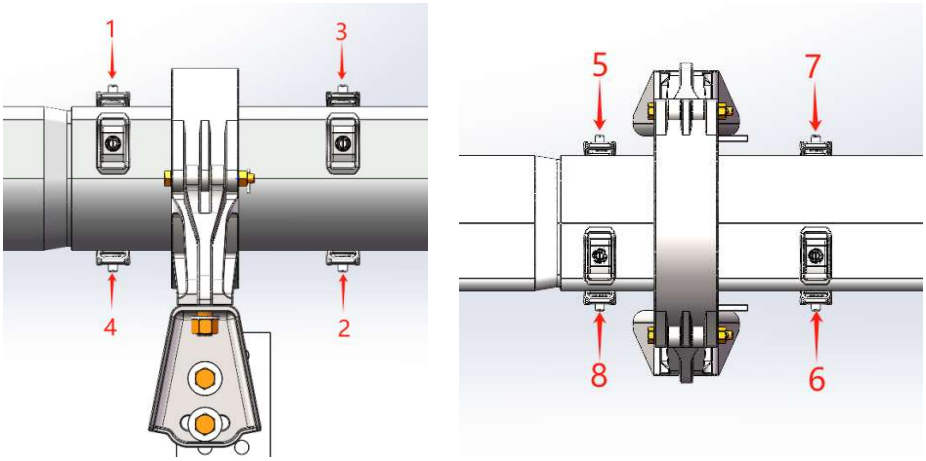


Figure 5.6.2-9: Schematic diagram of the installation sequence of the carriage bolts

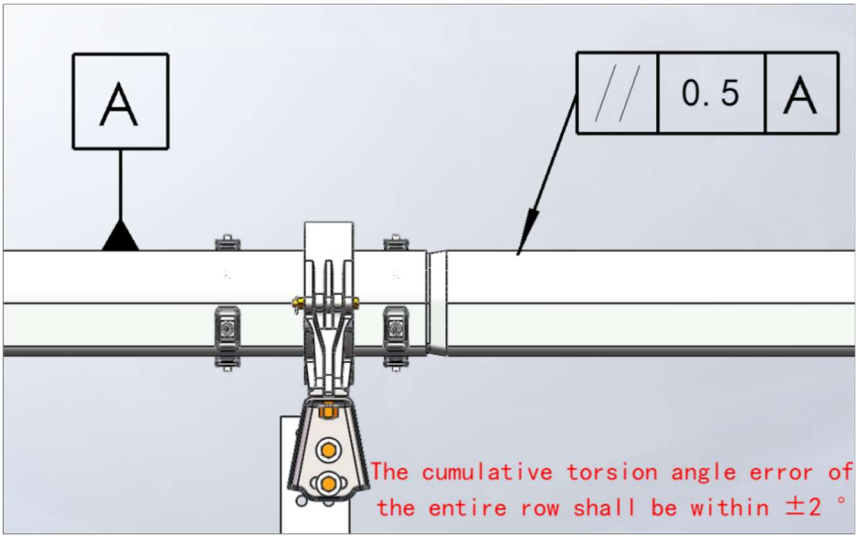


Figure 5.6.2-10: Schematic diagram of the parallelism of the torque tubes

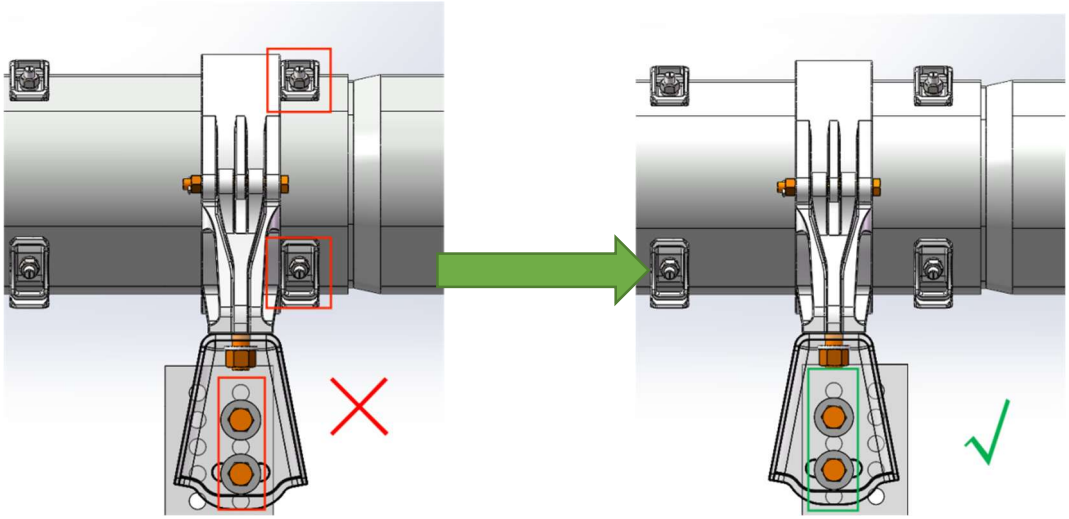


Figure 5.6.2-11: Adjust the positions of the mounting holes when the Bearing Housing and Carriage Bolt interfere.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.7 Installation of purlins (Step 7)

1. Prepare the installation parts, and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 16mm open wrench*2, leveling instrument, rubber hammer, tape and related equipment;
3. Purlins shall be installed in sequence from the middle of the bracket row to both sides. During installation, their angle shall be adjusted and secured.

Remarks:

① Purlin assemblies are installed onto the torque tube and secured, for the subsequent installation of PV modules,Mounting dimensions refer to **Appendix 2.5: Installation of purlins and PV modules**.

② Before and after tightening, it is necessary to ensure that the length direction of the purlin is perpendicular to the axial direction of the tube.

③ When tightening the U bolt, the bolts on both sides shall be tightened alternately and gradually. The bolts on one side shall not be fully tightened directly to the specified torque value, so as to prevent the U bolt from shifting to one side.

④ The outermost purlins at the north and south ends shall extend beyond the PV modules, and the modules at these positions shall be installed in the bolt holes on the inner side of the respective purlins.Refer to Figure 5.7-14.

Table 5.7-9: List of materials for tracker purlin installation

No.	Description	Specifications	Remark
1	Purlin	Refer to **Appendix 2.5: Installation of purlins and PV modules**	
2	U bolt	UM10*166*188*30	
3	Purlin pad block	BP095	
4	Purlin reinforcement gasket	- 3*22*35	
5	Flange nut	M10	Torque 25~32 N.m

info@esetsolar.com +86 516-83152579 www.esetsolar.com

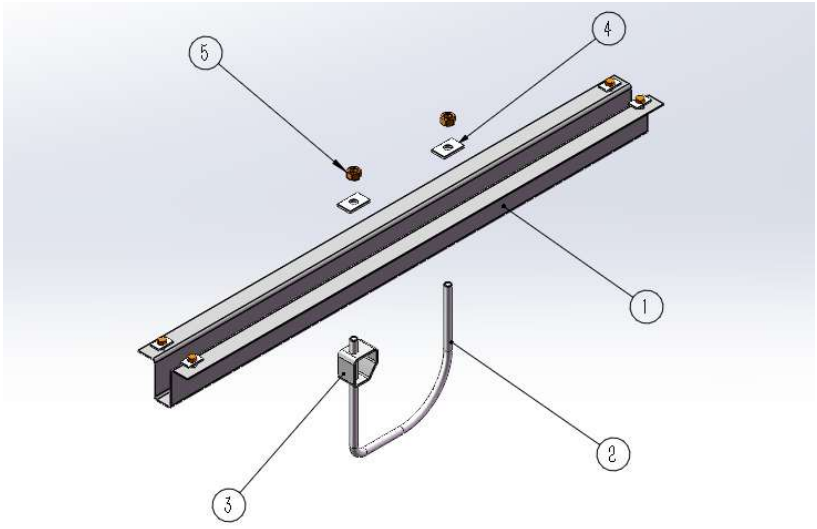
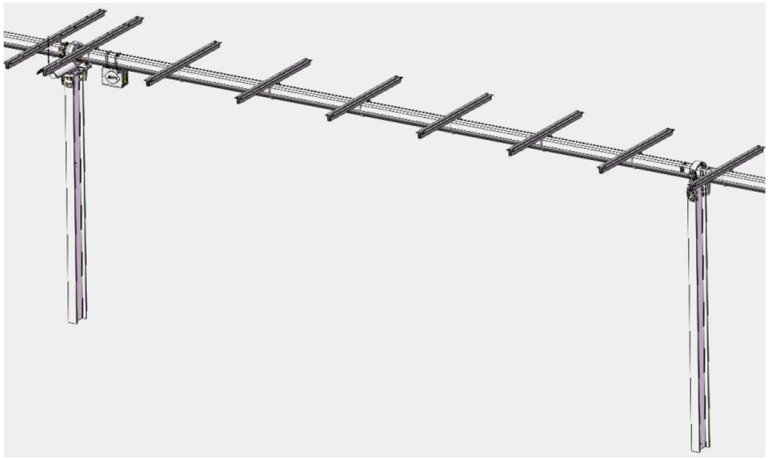


Figure 5.7-10: Installation and explosion diagram of purlin system



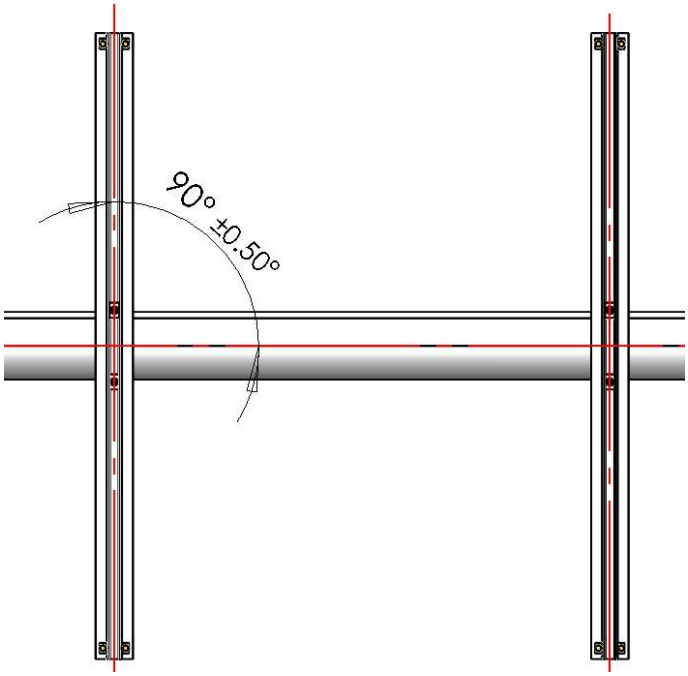
Figure 5.7-11: Pre-installation drawing of purlin system



1. The picture does not mean the real quantity of purlins.
2. Purlins shall be installed in sequence from the middle of the bracket row to both sides. During installation, their angle shall be adjusted and secured.

Figure 5.7-12: Installation drawing of purlin assemblies

info@esetsolar.com +86 516-83152579 www.esetsolar.com



1. Before and after tightening,it is necessary to ensure that the length direction the purlin is perpendicular to the axial direction of the tube, the permissible tolerance range is as shown in Figure 5.7-13.

Figure 5.7-13: Purlin installation diagram

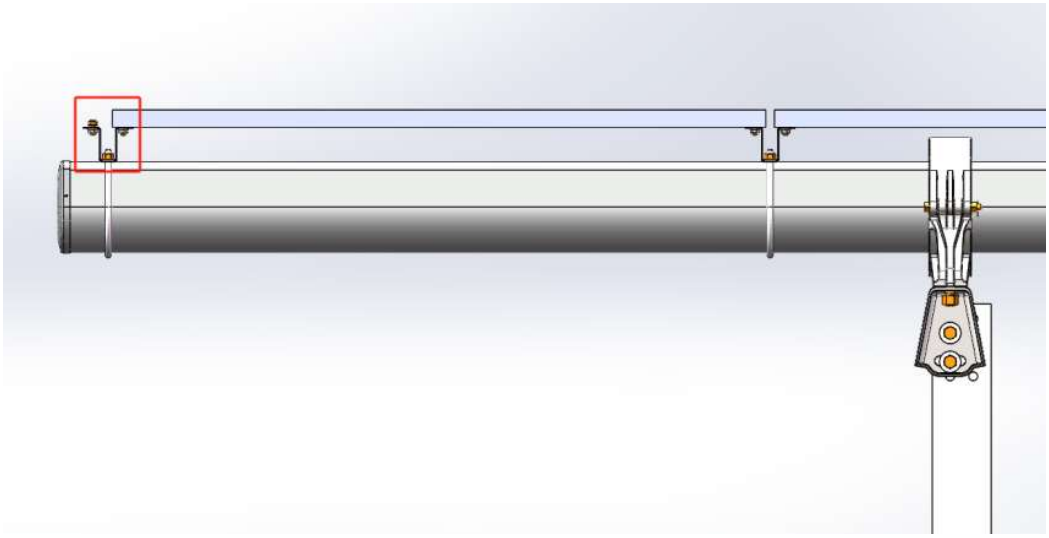
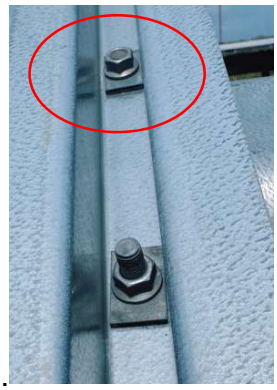


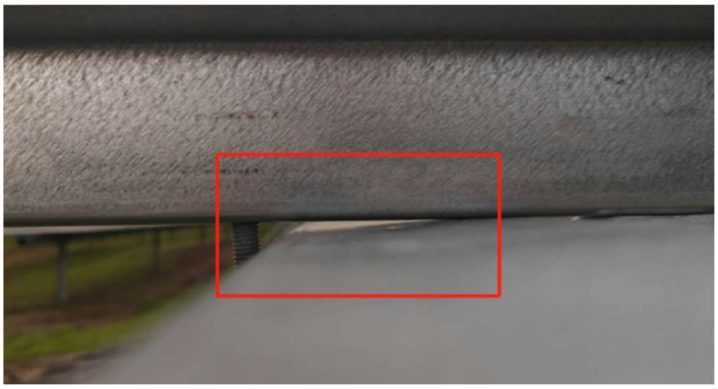
Figure 5.7-14: End purlin installation diagram

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Troubleshooting:

These bolt fastening methods shown in the images below are not permitted, as they reduce the contact area between the purlin and the torque tube (as illustrated in the image below), which in turn weakens the overall structural strength of the tracker.





Adjustment Methods:

For trackers without PV modules installed:

Use an electric wrench to adjust the nuts on the U-bolts. After adjustment, the exposed thread depth on both nuts should be greater than 2 turns.

The image below shows a correct installation example:



info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.8 Installation of the dampers (Step 8)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: 18mm open wrench * 2, 21mm open wrench * 2, leveling instrument, needle-nose pliers, and related supporting equipment
3. Install the damper lower fixture on the damper post; install the damper hoop on the torque tube, and tighten the nut by hand first; install the damper between the damper hoop and lower fixture, then fix the bolt with a pin (see Figure 5.8-6、5.8-7)

Remarks:☆ When installing dampers: the large diameter end is on the upper side.

① ☆ The dampers are symmetrically installed.

② ☆ Note that the fixing positions of the damper lower fixture differ between the damper and damper-S. Relevant requirements shall refer to Figure 5.8-3.

③ ☆ When installing dampers at positions other than the connection nodes, prioritize tightening the bolts at the damper lower fixture first. Then, confirm the installation position of the damper hoop based on the length of the damper upon delivery, and complete the installation after confirming the position (refer to Figure 5.8-4).

④ At the torque tube connection nodes, the installation position of the damper refers to Figure 5.8-5.

⑤ ☆ Insert the cotter pin into the hole of the bolt, split the pin and the angle of the pin is bigger than 180°, as shown in Figure 5.8-6.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Table 5.8-1: Materials required for damper installation

NO.	Description	Specifications	Remarks
1	Damper post	Refer to **Appendix 5: BOM List**	
2	Damper fixture	BP168	
3	Round tube of damper	Φ20*2.0	
4	Damper	/	
5	U Bolt	UM12*168*195*35	Torque 45~55 N.m
6	Long Purlin pad block	BP095L	
7	Damper lower fixture	BP093	
8	Hex bolt	M12 * 40 * 25 (1 double stack self-lock gasket 1 nut)	Torque 58~78 N.m
9	Hex bolt	M14 * 80 * 25 (1 slot nut 1 cotter pin 4 * 32)	

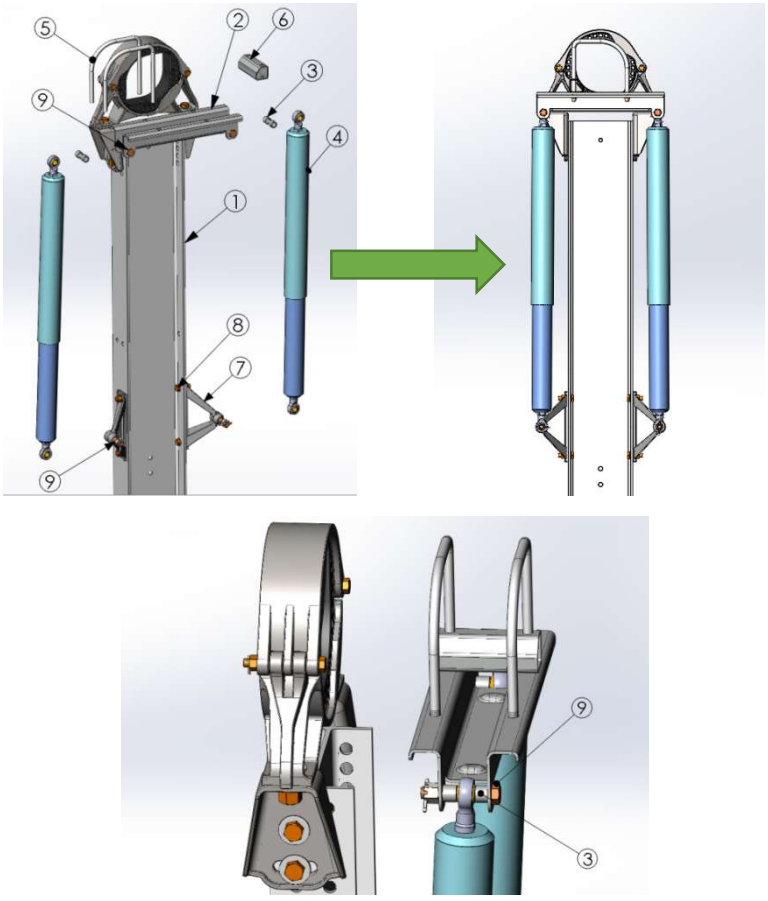


Figure 5.8-2: Schematic diagram of the damper installation

info@esetsolar.com +86 516-83152579 www.esetsolar.com

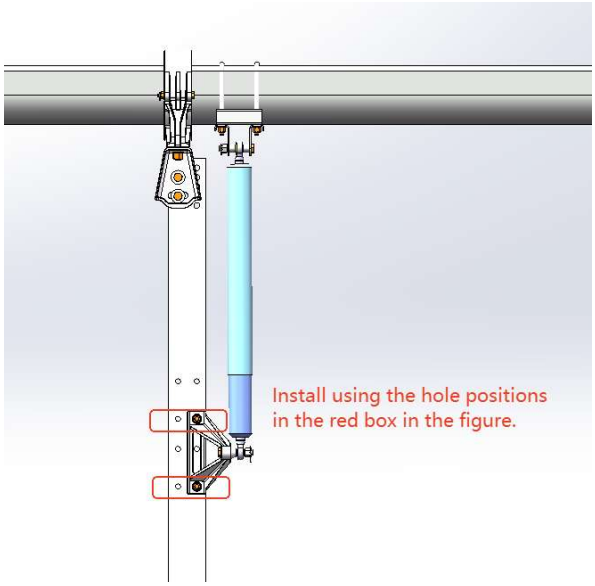


Figure 5.8-3: Schematic Showing the Differences in Installation Positions of the Damper Lower Fixture

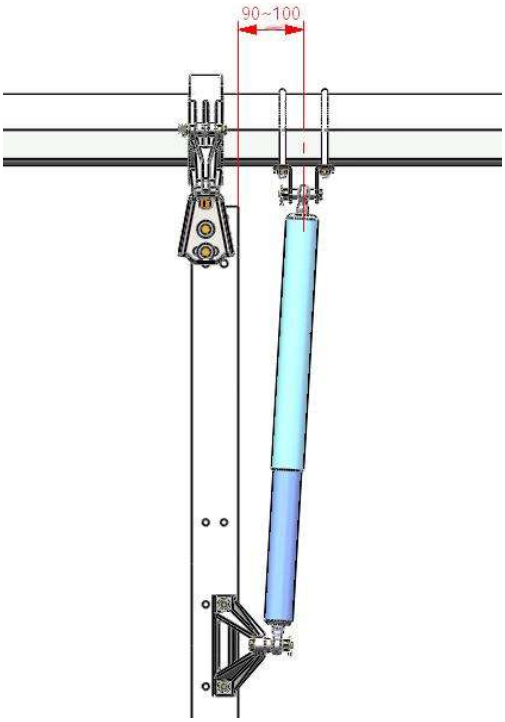
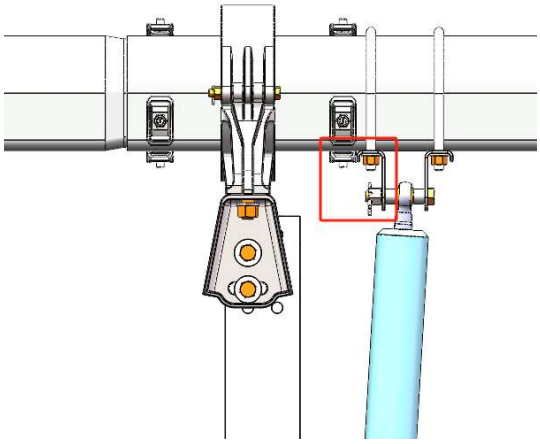


Figure 5.8-4: Schematic Diagram of Damper Installation at Positions Other Than the Connection Node of the Torque Tube

info@esetsolar.com +86 516-83152579 www.esetsolar.com



1. The edge of the damper fixture shall fit closely against the edge of the carriage bolt.

Figure 5.8-5: Schematic Diagram of Damper Installation at the Connection Node of the Torque tube

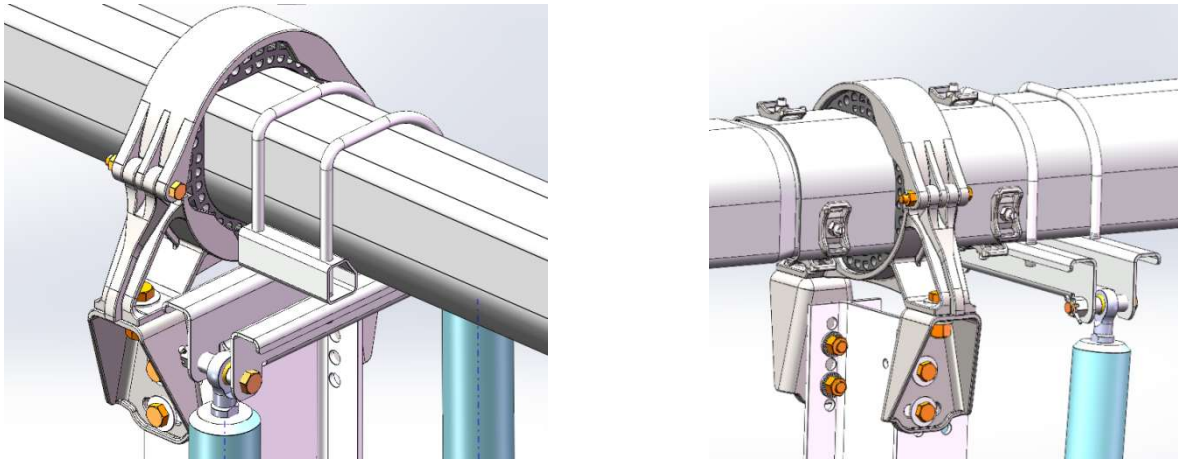


Figure 5.8-6: Diagram of damper hoop installation

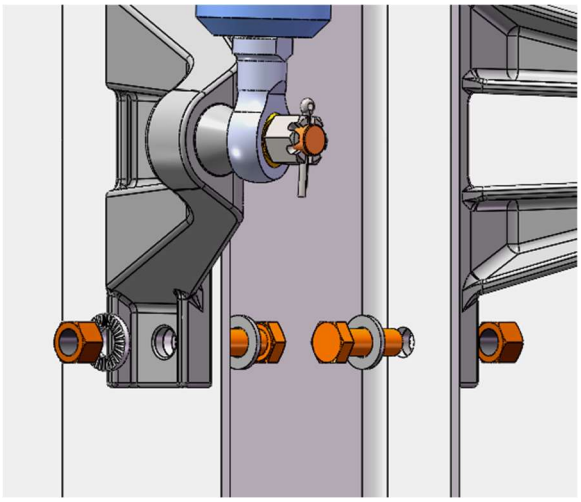


Figure 5.8-7: The connection diagram of the damper lower fixture

info@esetsolar.com +86 516-83152579 www.esetsolar.com

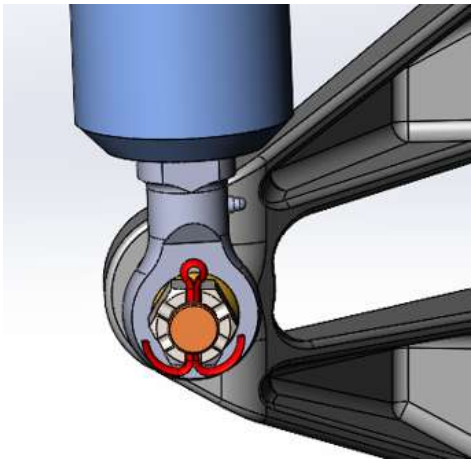


Figure 5.8-8: Drawing of cotter pin installation

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.9 Installation of TCU and wireless receiver (Step 9)

5.9.1 Installation of TCU

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents.
2. Prepare the installation tools: 6mm Allen wrench, and related supporting equipment
3. First, install the TCU hoop onto the torque tube, and then fasten the M8 bolts on the hoop to fix the TCU onto the torque tube.
4. The distance between the center of the TCU and the center of reducer should be controlled at 600 mm.

Remarks:

① Take waterproof, moisture-proof and other protection measures before the installation of the TCU;

② When installing the TCU, please pay attention to the label on the back of the TCU. The channel address has been pre-configured before the factory. During installation, ensure that the code on the TCU corresponds with the code marked in the Layout drawing；

③ The control box（TCU）is usually installed on the north side of the reducer, the 'N' symbol on the shell points to the north;

④ After the installation, fix the motor cable on the motor; the length of the cable should be slightly longer considering the rotation of the tracker.

Table 5.9-1: Materials required for installation of TCU

NO.	Description	Specifications	Remarks
1	Controller box（TCU）	Refer to **Appendix 5: BOM List**	
2	Hoop		

5.9.2 Installation of the wireless receiver

1. For the installation method of the wireless receiver, refer to the **Appendix 2.4 Installation of the wireless receiver.**

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.10 Installation of PV modules (Step 10)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: 10mm socket wrench, and related supporting equipment
3. Locate and tighten purlins. The interval from the center of the reducer to the purlin is L1, the interval from the purlin to the end of the torque tube is L2, and the interval between adjacent purlins is L3 and L4, refer to **Appendix 2.5.2: Installation of PV modules**. Screw the bolts on by hand first, and then tighten them with a power tool to the specified torque value when installing PV modules later (refer to Appendix 1)
4. Install the power supply module directly above the reducer, with their fixing purlins located on the north and south sides of the reducer respectively
5. Taking the reducer as the starting point, install the PV modules one by one from the center to both sides, with the spacing between the modules on both sides of the reducer as L4, the spacing between the end of the torque tube and the unilateral module as L5, and the spacing between the two PV modules as L6, refer to **Appendix 2.5.2: Installation of PV modules**.

Remarks:

① ☆ Installation method ： 2 people lift and 1 people install, bolted with hand first and then tighten them with electric wrench to the corresponding torque later.

② ☆ When fix the PV modules, ensure that the square gasket is installed on the PV module side . (see Figure 5.10-4)

③ ☆ The installation direction of the positive and negative poles of the PV module is "the positive pole is in the south".

④ ☆Refer to the Project Corresponding Installation Documents for the specific installation dimensions of L1~L7,refer to **Appendix 2.5.2: Installation of PV modules.**

⑤ After installation of PV modules, the tracker should be rotated to the 60° (wind protection angle), as shown in Figure 5.10-5.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Table 5.10-1: List of materials required for the installation of PV modules

NO.	Description	Specifications	Remarks
1	PV modules	/	
2	Stainless Steel Hex Bolt	M6 * 14 * 14 (1 piercing washer 1 big flange nut)	Torque 8 N.m
3	Stainless Steel Hex Bolt	M8 * 16 *16 (1 piercing washer 1 big flange nut)	Torque 16 N.m

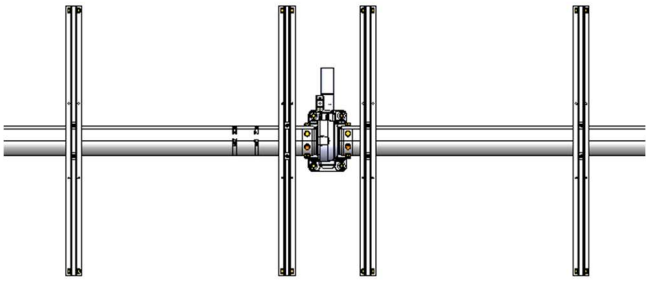


Figure 5.10-2: Schematic diagram of power supply module

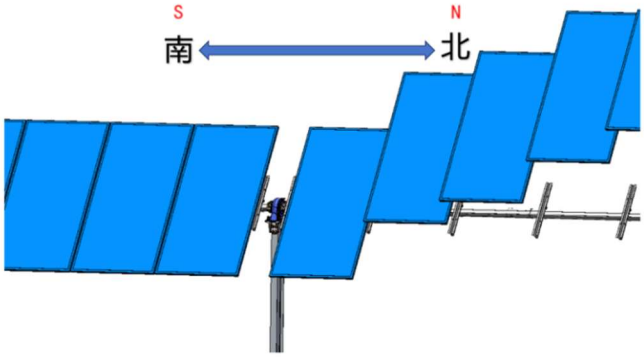


Figure 5.10-3: Schematic diagram of PV modules

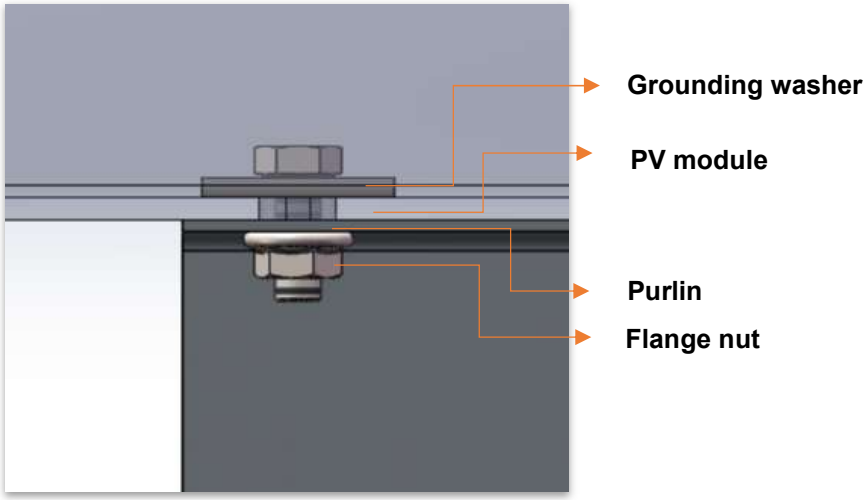


Figure 5.10-4: Installation diagram of PV module installation bolts

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.11 Wiring of PV system (Step 11)

General rules for Wiring routing

1. The cable wiring should be beautiful and neat, the fixing should be firm, the reserved length should be moderate, and the cable should not be stressed when the bracket rotates.
2. Cables exposed to sunlight require a bellows.
3. When burying wiring, the buried depth is required to be not less than 0.8m.
4. Cables must be protected when entering and exiting the cable tray, conduit, and holes through metal partitions.
5. Workers must carefully and strictly adhere to the module manufacturer safety information for the proper installation, maintenance, and handling of the modules. All module manufacturer instructions and requirements must be read and understood before attempting to install, wire, and operate and/or service the module and accompanying electrical equipment.
6. This procedure provides simplified, schematic examples of recommended wire management configurations to avoid contact with tracker mounting components. All wire management must comply with IEC standards and local policy requirements for maintaining wires (service loops, cable trays, raceways, and so on).
7. Do not attach cable trays or shields to the underside of the panels. The maximum weight capacity attached to the back of modules is not to exceed 0.8 kg per meter.
8. ☆ For the arrangement and details of solar cables in the tracker, refer to the document ***SAU1004-G000-MQA-&ETL-00002 SOLAR CABLE ARRANGEMENT AND DETAILS IN TRACKER***

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.12 Installation of grounding wire (Step 12)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: socket wrench, cross screwdriver, and related supporting equipment
3. Attach one end of the yellow-green ground wire to the web of the column and the other end to the Grounding Clamp, then tighten the nuts at both connections,refer to Figure 5.12-4.
4. All Grounding Clamps are installed next to the first column at both the north and south ends, regardless of whether there is a damper at this location, refer to Figure 5.12-1.
5. The ground wire of TCU is integrated in motor wire, when motor wire is connected with motor, grounding function has equipped directly.

Table 5.12-1 :List of Materials Required for the Installation of Yellow-Green Ground Wires

NO.	Description	Specifications	Remarks
1	Grounding Cable	S156	
2	thin nut	M12	
3	Hex bolt	M10 * 25 (1 flat washer, 1 nut)	Secure the Grounding Cable

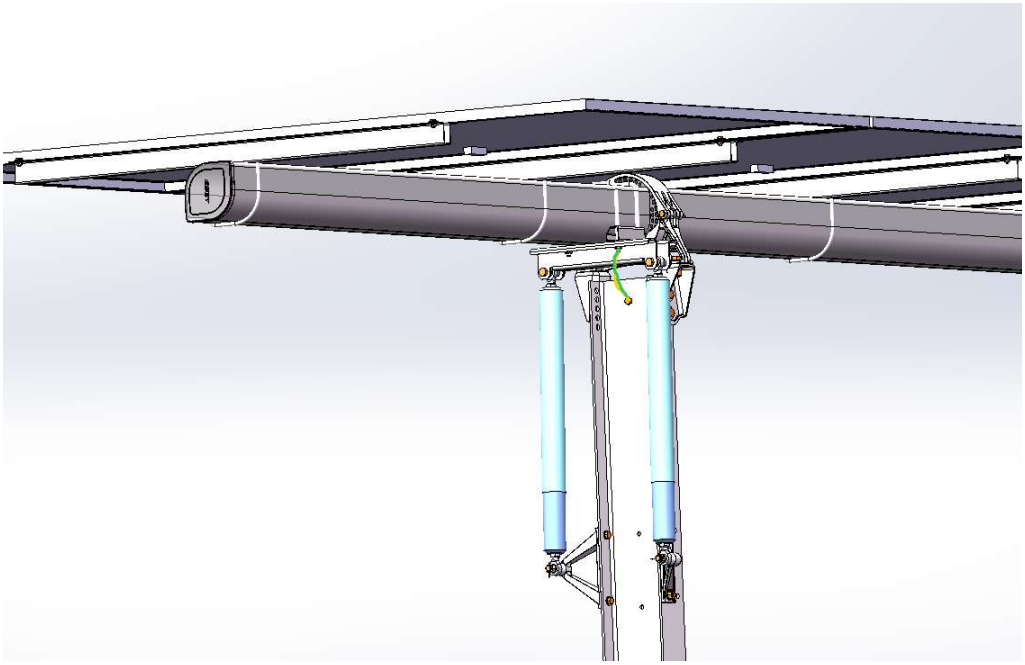


Figure 5.12-2:Installation schematic for tracker grounding clamp

info@esetsolar.com +86 516-83152579 www.esetsolar.com

5.13 Installation of torque tube cover (Step 13)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents；
2. After the torque tube is installed, finally install the cover of the torque tube at both ends, as shown in Figure 5.13-2.

Table 5.13-1: Materials required for torque tube cover installation

NO.	Description	Specifications for S156	Remarks
1	Torque tube cover	BP098	

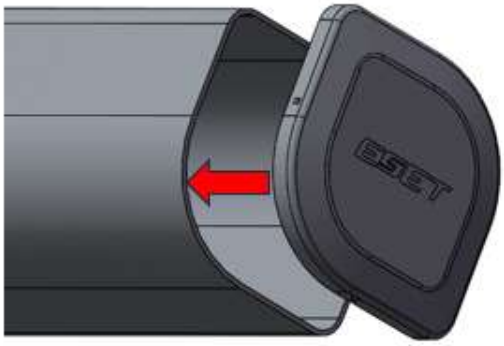


Figure 5.13-2: Installation diagram of torque tube cover

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Appendix 1: Description of the tightening torque of the bolts

Bolt strength grade	Yield strength	Bolt nominal diameter / mm			
		6	8	10	12
	N/mm^2	Tightening torque of M / N.m			
4.6	240	4 ~ 5	10 ~ 12	20 ~ 25	36 ~ 45
6.8	300	5 ~ 7	12 ~ 15	25 ~ 32	45 ~ 55
8.8	480	7 ~ 9	17 ~ 23	33 ~ 45	58 ~ 78
Bolt strength grade	Yield strength	Bolt nominal diameter / mm			
		14	16	18	20
	N/mm^2	Tightening torque of M / N.m			
4.6	240	55 ~ 70	90 ~ 110	120 ~ 150	170 ~ 210
6.8	300	70 ~ 90	110 ~ 140	150 ~ 190	210 ~ 270
8.8	480	93 ~ 124	145 ~ 193	199 ~ 264	282 ~ 376
Bolt strength grade	Yield strength	Bolt nominal diameter / mm			
		22	24	27	30
	N/mm^2	Tightening torque of M / N.m			
4.6	240	230 ~ 290	300 ~ 377	450 ~ 530	540 ~ 680
6.8	300	290 ~ 350	370 ~ 450	550 ~ 700	680 ~ 850
8.8	480	384 ~ 512	488 ~ 6 50	714 ~ 952	969 ~ 1293

Notes：

1. **After the tracker is installed, the bolts must be tested one by one with a torque wrench to see if the tightening torque value is reached and marked with a line.**
2. **If there is damage to the zinc layer on the surface after the bolt is tightened, surface rust prevention is required.**
3. **After three months of normal operation of the tracker, all the bolts connecting the torque tube need to be inspected and the relevant bolts need to be tightened for a second time.**

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Appendix 2: Project-specific installation dimensions

Appendix 2.1: Installation of posts

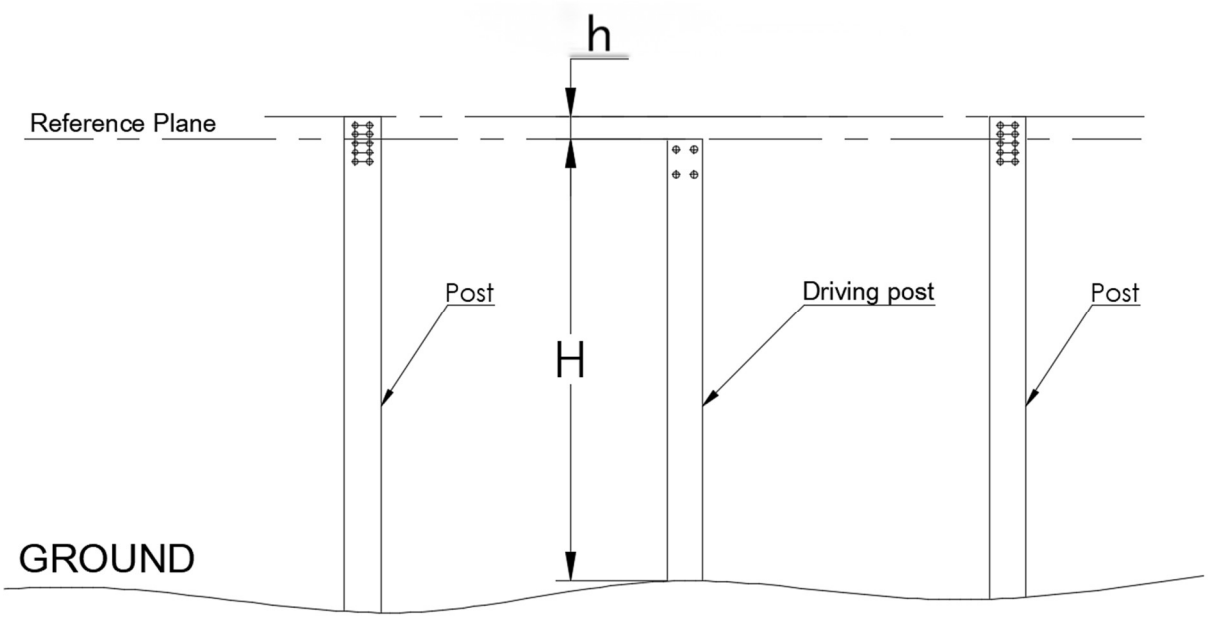
① The installation direction of posts is the north and south;

② Refer to Table a2.1-1 to Figure a2.10 for foundation height;

③ Refer to Table a2.1-2 to Figure a2.1-10 for post installation location and spacing.

④ *Take the following model as an example for reference.

Table a2.1-1: Schematic chart of foundation depth and height from the ground



No.	1x28-Exterior	1x42-Exterior	1x56-Exterior	1x70-Exterior
	5 Inch	6 Inch	7 Inch	5 Inch
H	1150±100 (mm)	1134±100 (mm)	1124±100 (mm)	1150±100 (mm)
h	46 (mm)	62 (mm)	72 (mm)	46 (mm)

Remark:

* With **H** as the reference, the remaining dimensions shall move accordingly.

* Taking **h** as the reference dimension for the ground clearance of the top surface of the Driving Post, this height dimension will vary within ±100mm with slope variations, and the remaining

dimensions shall be adjusted accordingly.

Table a2.1-3: Materials required for the installation of 1x28 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 1	W8*10	QZ1-3050	1	
2	Non-driving post 1	W8*10	FZ1-3096	3	
3	Damper post 1	W8*10	NZ1-3096	2	

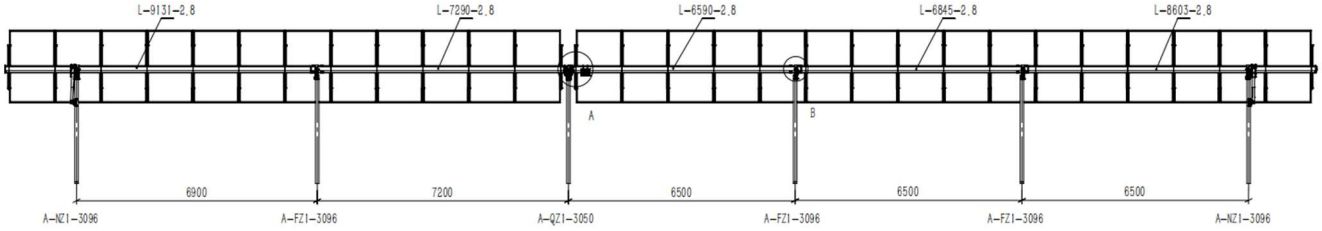


Figure a2.1-4: Schematic diagram of the installation position of 1P*28 **Exterior Tracker** posts with different part number.

Table a2.1-5: Materials required for the installation of 1x42 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 2	W8*13	QZ2-3034	1	
2	Non-driving post 1	W8*10	FZ1-3096	6	
3	Damper post 1	W8*10	NZ1-3096	2	

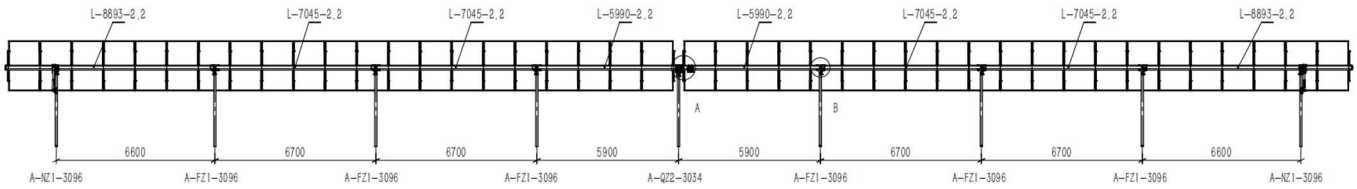


Figure a2.1-6: Schematic diagram of the installation position of 1P*42 **Exterior Tracker** posts with different part number.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Table a2.1-7: Materials required for the installation of 1x56 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 3	W8*15	QZ3-3024	1	
2	Non-driving post 1	W8*10	FZ1-3096	6	
3	Non-driving post 2	W8*13	FZ2-3096	2	
4	Damper post 1	W8*10	NZ1-3096	2	

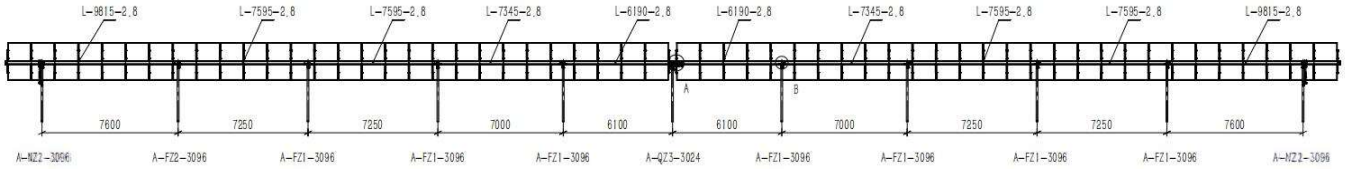


Figure a2.1-8: Schematic diagram of the installation position of 1P*56 **Exterior Tracker** posts with different part number.

Table a2.1-9 Materials required for the installation of 1x70 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 4	W8*13	QZ4-3050	2	
2	Non-driving post 1	W8*10	FZ1-3096	9	
3	Damper post 1	W8*10	NZ1-3096	3	

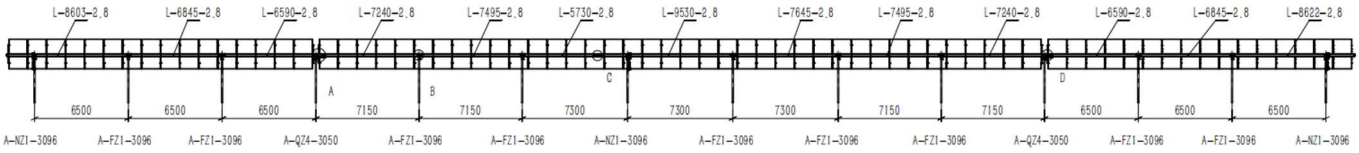


Figure a2.1-10: Schematic diagram of the installation position of 1P*70 **Exterior Tracker** posts with different part number.

For detail information, please check Pile foundation construction guidance.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Appendix 2.2: Torque tube installation process

① See Figure a2.2-1 to a2.2-22 for details of torque tube installation locations with different part number;

Table a2.2-1: Materials required for the **Torque Tube** installation of 1P*28 **Exterior Tracker**

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A1	S156x2.8	L-7290-2.8	1	
2	Torque tube A2	S156x2.8	L-6590-2.8	1	
3	Torque tubeB1	S156x2.8	L-6845-2.8	1	
4	Torque tube C1	S156x2.8	L-8603-2.8	1	
5	Torque tube C2	S156x2.8	L-9131-2.8	1	

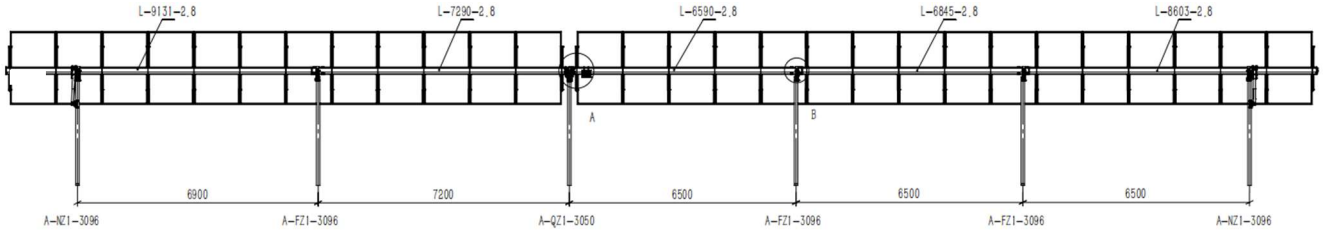


Figure a2.2-2: Schematic diagram of 1P*28 **Exterior Tracker** torque tube mounting position with different part number.

Table a2.2-3: Materials required for the **Torque Tube** installation of 1P*42 **Exterior Tracker**.

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A5	S156x2.2	L-5990-2.2	2	
2	Torque tube B6	S156x2.2	L-7045-2.2	4	
3	Torque tube C5	S156x2.2	L-8893-2.2	2	

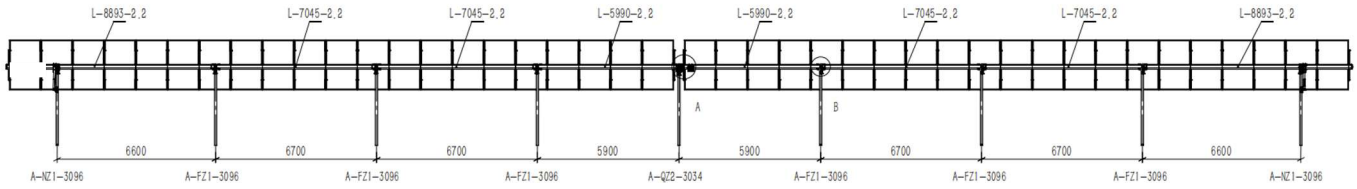


Figure a2.2-4: Schematic diagram of 1P*42 **Exterior Tracker** torque tube mounting position with different part number.

Table a2.2-5: Materials required for the **Torque Tube** installation of 1P*56 **Exterior Tracker**.

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A3	S156x2.8	L-6190-2.8	2	
2	Torque tube B2	S156x2.8	L-7345-2.8	2	
3	Torque tube B3	S156x2.8	L-7595-2.8	4	
4	Torque tube C3	S156x2.8	L-9815-2.8	2	

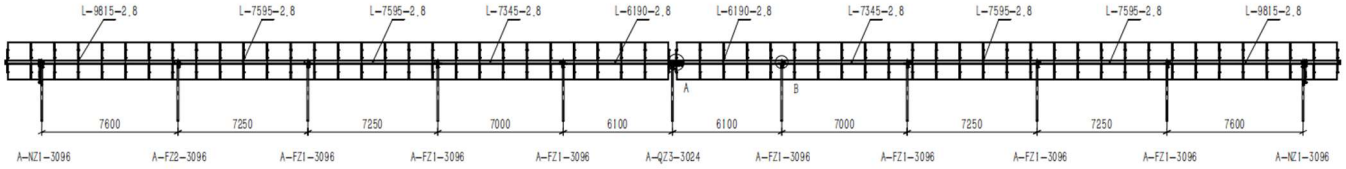


Figure a2.2-6: Schematic diagram of 1P*56 **Exterior Tracker** torque tube mounting position with different part number.

Table a2.2-7: Materials required for the **Torque Tube** installation of 1P*70 **Exterior Tracker**.

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A2	S156x2.8	L-6590-2.8	2	
2	Torque tube A4	S156x2.8	L-7240-2.8	2	
3	Torque tube B1	S156x2.8	L-6845-2.8	2	
4	Torque tube B4	S156x2.8	L-7495-2.8	2	
5	Torque tube B5	S156x2.8	L-7645-2.8	1	
7	Torque tube C4	S156x2.8	L-8613-2.8	2	
8	Torque tube D1	S156x2.8	L-5730-2.8	1	
9	Torque tube D2	S156x2.8	L-9530-2.8	1	

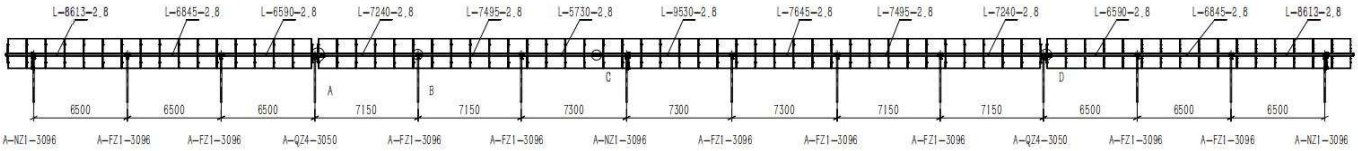


Figure a2.2-8: Schematic diagram of 1P*70 **Exterior Tracker** torque tube mounting position with different part number.

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Appendix 2.3: Installation of TCU and wireless receiver

Appendix 2.3.1 Installation of TCU

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents.
2. Prepare the installation tools: 6mm Allen wrench, and related supporting equipment
3. First, install the TCU hoop onto the torque tube, and then fasten the M8 bolts on the hoop to fix the TCU onto the torque tube.
4. The distance between the center of the TCU and the center of reducer should be controlled at 600 mm.

Remarks:

① Take waterproof, moisture-proof and other protection measures before the installation of the TCU;

② The control box（TCU）is usually installed on the north side of the reducer, the 'N' symbol on the shell points to the north;

③ After the installation, fix the cable of the motor on the post; the length of the cable should be slightly longer considering the rotation of the tracker.

Table a2.3.1-1: Materials required for installation of TCU

NO.	Description	Specifications	Remarks
1	Controller box（TCU）	/	
2	Hoop	L=530	

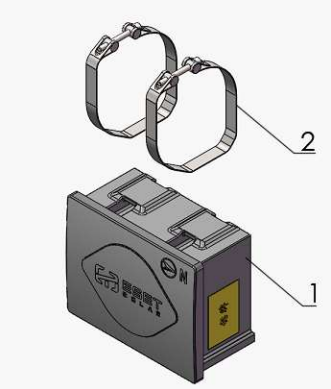


Figure a2.3.1-2: Installation and explosion diagram of the TCU-M

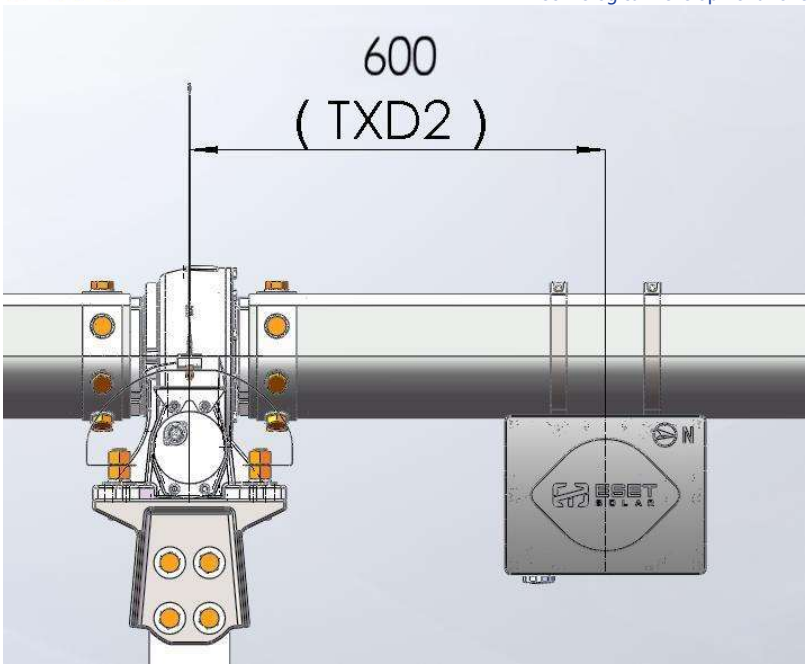


Figure a2.3.1-3: Diagram of installation position of TCU-M

Table a2.3.1-4: Materials required for installation of TCU-S

NO.	Description	Specifications	Remarks
1	Controller box（TCU）	/	
2	Hoop	L=530	

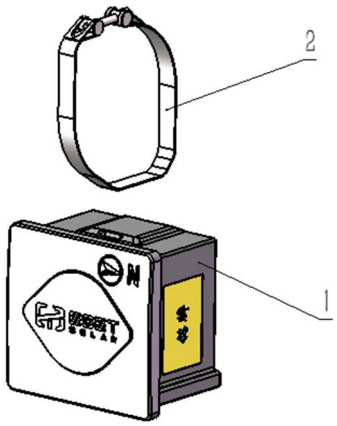


Figure a2.5.1-5: Installation and explosion diagram of the TCU-S

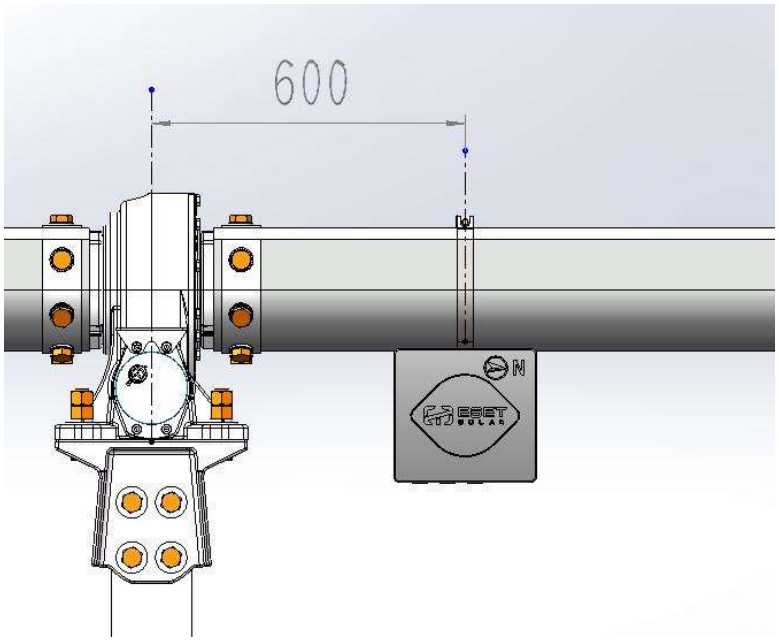


Figure a2.5.1-6: Diagram of installation position of TCU-S

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Appendix 2.3.3 Installation of the wireless receiver

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Place the wireless receiver on the wireless receiver bracket (on the slew drive cover) and fix it with M5 * 15 bolts

Table a2.3.3-1: List of materials required for the installation of the wireless receiver

NO.	Description	Specifications	Remarks
1	Wireless receiver bracket2	-2x55x90	
2	Wireless receiver	/	
3	Slew drive cover	-1x120x260	

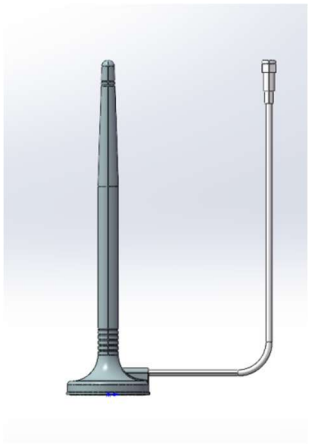


Figure a2.3.3-2: Schematic diagram of the wireless receiver

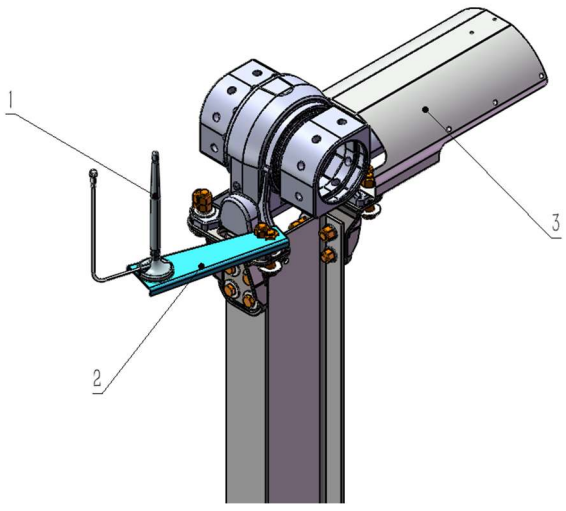
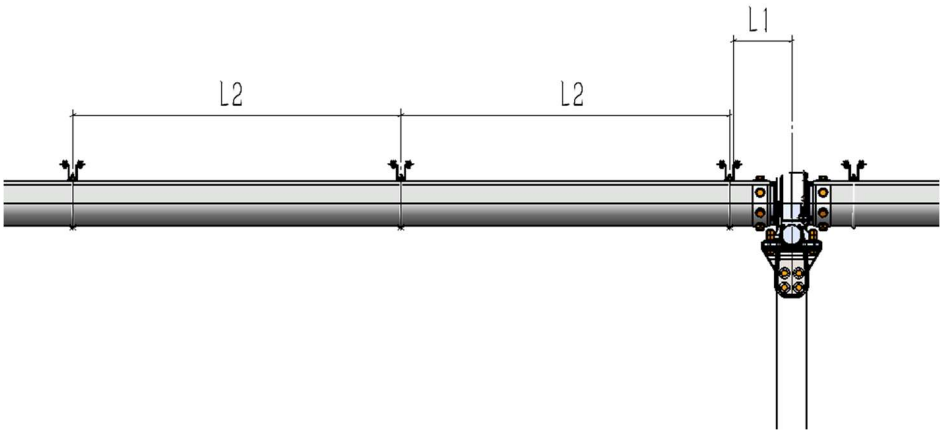


Figure a2.3.3-3: Installation diagram of the wireless receiver

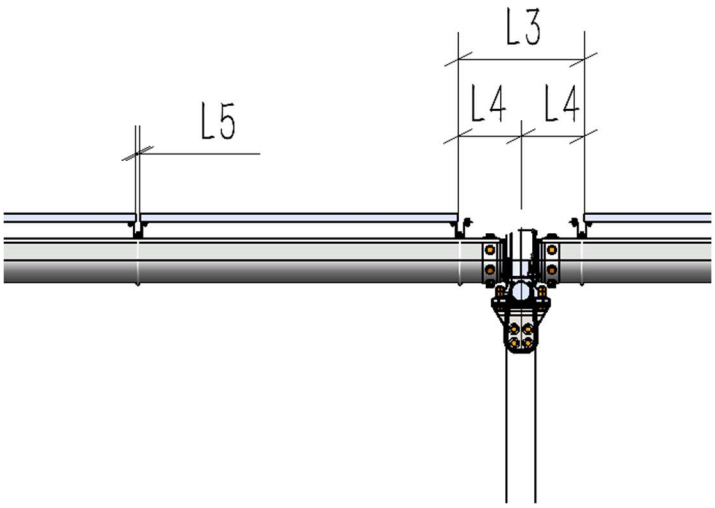
Appendix 2.4: Installation of purlins and PV modules

Appendix 2.4.1 Installation of purlins



Pos.	Dimension(mm)
L1	204
L2	1318

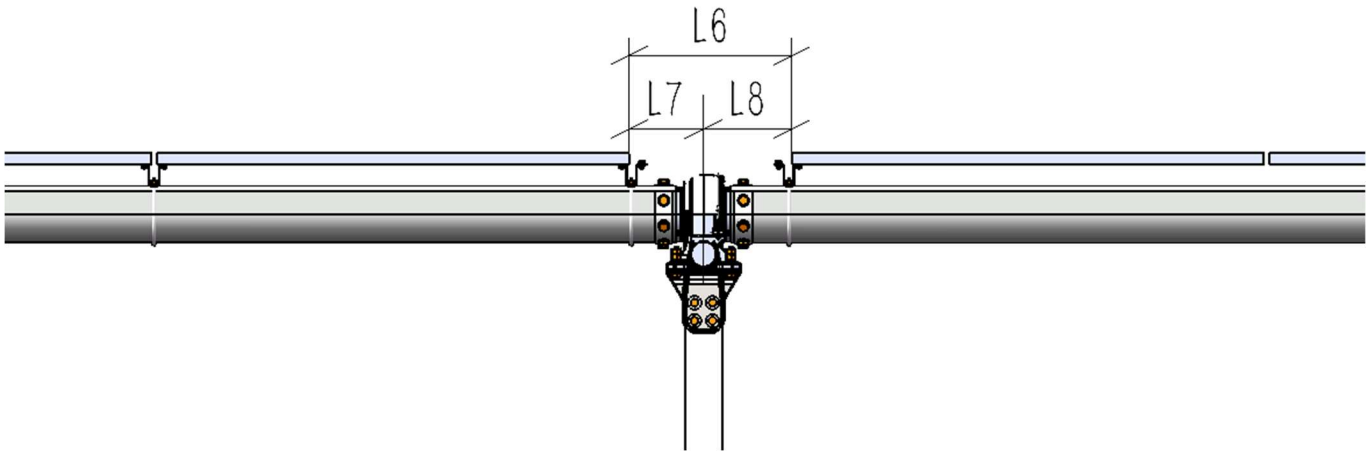
Figure a2.4.1-1: Schematic diagram of purlin installation distance(1x28，1x42，1x56，1x70)



Pos.	Dimension(mm)
L3	450
L4	225
L5	15

Figure a2.4.1-2: Schematic diagram of the installation distance of PV modules (1x28，1x42，1x56，1x70-TCU-M)

info@esetsolar.com +86 516-83152579 www.esetsolar.com



Pos.	Dimension(mm)
L6	450
L7	216
L8	234

Figure a2.4.1-2: Schematic diagram of the installation distance of PV modules (1x70-TCU-S)

Table a2.4.1-3: List of materials for 1x28 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.8	30	LT1-1450	
2	U bolt	UM10*166*188*30	30	/	
3	Purlin pad block	28*43.5*42	30	/	
4	Purlin reinforcement gasket	- 3*22*35	60	/	
5	Flange nut	M10	60	/	Torque 25-32 N.m

info@esetsolar.com +86 516-83152579 www.esetsolar.com

Table a2.4.1-4: List of materials for 1x42 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.8	44	LT1-1450	
2	U bolt	UM10*166*188*30	44	/	
3	Purlin pad block	28*43.5*42	44	/	
4	Purlin reinforcement gasket	- 3*22*35	88	/	
5	Flange nut	M10	88	/	Torque 25-32 N.m

Table a2.4.1-5: List of materials for 1x56 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.8	58	LT1-1450	
2	U bolt	UM10*166*188*30	58	/	
3	Purlin pad block	28*43.5*42	58	/	
4	Purlin reinforcement gasket	- 3*22*35	116	/	
5	Flange nut	M10	116	/	Torque 25-32 N.m

Table a2.4.1-6: List of materials for 1x70 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.6	73	LT1-1450	
2	U bolt	UM10*166*188*30	73	/	
3	Purlin pad block	28*43.5*42	73	/	
4	Purlin reinforcement gasket	- 3*22*35	146	/	
5	Flange nut	M10	146	/	Torque 25-32 N.m

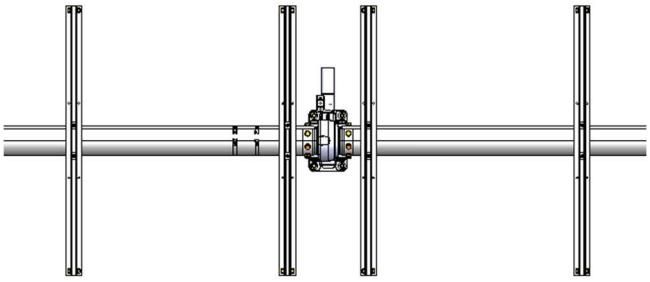
info@esetsolar.com +86 516-83152579 www.esetsolar.com

Appendix 2.4.2 Installation of PV modules

1. All PV modules shall be fastened with four bolts during installation.

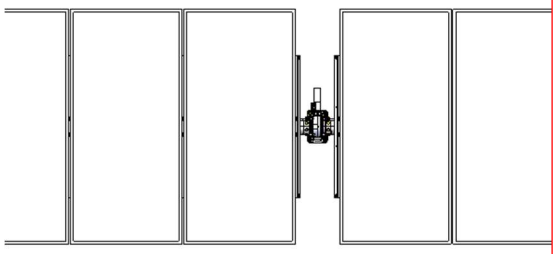
Table a2.4.2-1: List of materials required for the installation of PV modules

NO.	Description	Specifications	Remarks	Remarks 2
1	PV modules	/		Refer to **Appendix 5: BOM List**
2	Hex bolt	M6 * 14 * 14 (1 Grounding washer 1 big flange nut)	Torque 8 N.m	
3	Hex bolt	M8 * 16 *16 (1 Grounding washer 1 big flange nut)	Torque 16 N.m	



1. The installation position of the power supply module is above the reducer and on the opposite side of the drive motor.

Figure a2.4-16: Schematic diagram of power supply module



1. The end faces of the power supply module shall be flush with those of the PV modules.

Figure a2.6-17: Schematic diagram of power supply module

info@esetsolar.com +86 516-83152579 www.esetsolar.com